

Options Trading Costs Are Lower Than You Think[☆]

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Abstract

Conventional estimates of costs of taking liquidity in options markets are large. Nonetheless, options trading volume is high. We resolve this puzzle by showing that options price changes are predictable at high frequency and many traders time executions by buying (selling) when the option fair value is close to the ask (bid). Effective spreads of traders who time executions are less than 40% of the size of conventional measures, and the overall average effective spread is one-quarter smaller than conventional estimates. Price impact measures are also affected. These findings alter conclusions about the after-cost profitability of options trading strategies.

JEL Classification: G13, G14

Keywords: Execution timing, trading costs, effective spread, liquidity, equity options, algorithmic trading

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During our sample period of 2004-2015 quoted half-spreads of options on stocks in the S&P 500 index averaged 13 cents per share and 8.6% of the option price. Dollar (percentage) spreads were considerably wider for well in-the-money (out-of-the-money) options. Conventionally measured effective half-spreads averaged 10 cents per share or 6.8% of the option price, with dollar (percentage) spreads again much wider for well in-the-money (out-of-the-money) options. Given these quoted and effective spreads, the costs of taking liquidity in the options market seem high. But despite these apparently high costs, U.S. options are very actively traded. For example, during 2014 approximately 3.85 billion option contracts on equities and ETFs were traded, which translates to contracts on 385 billion shares and is equal to 23.79% of the 1,616 billion share trading volume in the U.S. equity markets.¹ It has been estimated that more than half of options trading is by institutional investors, most of whom are likely sufficiently sophisticated to care about transactions costs (Aite Group 2008). The fact that presumably sophisticated investors are trading so many options at such seemingly high costs is a puzzle.

While the costs of options market making can help explain why options spreads should be higher than the spreads of their underlying stocks (Battalio and Schultz 2011), a second puzzle is that existing theories are unable to explain the observed patterns of spreads. For example, the high dollar spreads of in-the-money (ITM) options and the relation between spreads and moneyness cannot be explained by hedge rebalancing costs incurred by options market makers, because hedges of well ITM options rarely need to be rebalanced. Similarly, the pattern cannot be explained by difficult to hedge gamma and vega risks that options market makers bear when they hold inventories of options, because well ITM options are not exposed to these risks. Through the put-call parity relation ITM calls (puts) have gamma and vega risks similar to those of their corresponding out-of-the-money (OTM) puts (calls), but much different spreads. The differences between the spreads of options and their underlying stocks also cannot be explained by differences in the adverse selection component of the spread unless informed traders are much more common in the options market than in the stock market and they choose to trade ITM options rather than ATM or OTM options that have more embedded leverage. Finally, option volume increased substantially while quoted spreads decreased little during our sample period,

¹ Option and stock volume data are from Options Clearing Corporation (OCC) and Securities Industry and Financial Markets Association (SIFMA) websites respectively.

which seems inconsistent with both theories in which reductions in trading costs increase trading volume and theories in which increases in trading volume lead to lower costs per unit.

This paper helps resolve these puzzles. We begin by using a sample of quote snapshots at one-minute frequency for all options on S&P 500 stocks from 2004-2015 to show that options prices are predictable at high frequency.² The difference between a Black-Scholes-Merton (BSM) model value based on recent implied volatility and the current options quote midpoint predicts changes in the quote midpoint. This predictability is enhanced using lagged changes in stock and options prices and limit order book information. The predictors are all publicly available, and we interpret the expected future midpoints computed using the models as estimates of options fair values. The average R^2 of 14% is remarkably high for predicting future intraday price changes with past public information. Out-of-sample R^2 s are also large, and the predictive coefficients are consistent across moneyness categories, groups of stocks, and time.

Options traders exploit this predictability in timing their executions. Executions at the bid price tend to occur when the estimate of the fair value (the expected future midpoint) is close to but less than the quoted ask price, and executions at the ask price tend to occur when it is close to but greater than the quoted bid price.³ Traders who exploit this predictability are able to take liquidity at low costs, as we explain next.

The effective spread is the difference between the transaction price and the “true value” (Goettler, Parlour, and Rajan 2005). As Bessembinder and Venkataraman (2010) explain, in empirical analyses the effective spread is computed using “an observable proxy for the true underlying value of [the] security.” Researchers often use the bid-ask midpoint as a proxy for the value and thus measure the effective half-spread as the difference between the transaction price and the bid-ask midpoint, adjusted for trade direction. If the timing of executions is uncorrelated with the errors in using the midpoint as a proxy, using it will result in unbiased estimates of average trading costs even though the errors resulting from the use of the proxy in place of the “true value” or fair value result in noisy estimates of the effective spread for individual trades. But if the timing of executions is correlated with the errors in using the midpoint as a proxy then the estimates of trading costs will be biased. This happens, for example, if executions are more

² We report results for non-S&P 500 stocks in the internet appendix.

³ Thus, this trade timing does not imply the existence of frequent profit or arbitrage opportunities. Almost all of it occurs within the quoted spread.

likely to occur after some traders have entered aggressive limit orders, or if some traders are slow to cancel stale limit orders.

Because executions at the ask (bid) price tend to occur when the estimates of options' fair values are close to the ask (bid) price, the difference between the option value and the bid-ask midpoint is large conditional on the occurrence of a trade, and correlated with the trade direction. For example, an execution at the ask price might occur when the estimate of fair value is two cents below the ask price but the quote midpoint is five cents below the ask price. In this case the effective half-spread based on the fair value is only two cents but the conventional estimate based on the midpoint would be five cents.

We estimate that large fraction of the trades in options on S&P 500 stocks, about 38.2%, exploit the high frequency predictability of options prices to time executions.⁴ For trades that display high-frequency trade timing ability, the effective half-spread that accounts for it by using the predicted future midpoint as the estimate of value is on average 2.5% of the option price, just 37.4% of the conventionally measured effective half-spread of 6.8% and 29.6% of the average quoted half-spread of 8.6%. This new measure of effective half-spreads for execution timers decreases from 4% of the option price at the beginning to 1.5% by the end of the sample period. This decrease in execution timers' spreads coincides with a decrease in market volatility and an increase in the share of trades that show execution timing ability. The estimates help resolve the puzzle of why options trading volume is so high despite the seemingly very high trading costs—trading costs of investors who use execution algorithms to time executions are much lower than the conventional estimates of costs. The investors who time executions likely include institutional investors who use either their own or their brokers' execution algorithms, and possibly some retail investors who use their brokers' execution algorithms. Because most execution timing is likely carried out using execution algorithms, we sometimes refer to the spread paid by execution timers as the algo spread.

The average effective spread of investors who do not time executions (the non-algo spread) is equal to the conventional estimate based on the midpoint. Thus, execution timing can

⁴ This estimate recognizes that half of untimed trades have positive timing by chance. We compute the estimate of 38.2% using the idea that for all timed trades the difference between the fair value and the midpoint, adjusted for trade direction, is positive. This immediately implies that any trade for which the adjusted difference is negative is an untimed trade. Then using the assumption that half of untimed trades display positive timing and half display negative timing, the fraction of untimed trades is twice the fraction of trades that display negative timing, and the fraction of timed trades is one minus two times the fraction that display negative timing.

serve as a discrimination mechanism resulting in different trading costs for investors who do and do not time executions. Averaging over trades by both timers and non-timers, the effective half-spread measure that account for trade timing is just 5.0%, or 26.3% less than the conventionally measured effective half-spread and 41.6% less than the average quoted half-spread. We estimate that the aggregate difference between the conventional measure of the costs of taking liquidity and our measure that accounts for execution timing is more than one billion dollars per year.

The execution timing that we document also helps resolve the puzzle of why dollar spreads of ITM options are so much larger than those of OTM options even though such options are not exposed to the gamma and vega risks that are sometimes offered as an explanation for high options bid-ask spreads. Indeed, the difference between spreads for OTM and ITM options drops by a factor of two after accounting for execution timing. Furthermore, trading costs for execution timers are similar for all moneyness categories.

We also provide measures of price impact that take account of execution timing. Conventional measures of price impact are estimated from the difference between the midpoints a short time after the trade and at the time of the trade. We replace the pre-trade midpoint with our estimate of value, the expected future midpoint, and obtain estimates of price impact that are less than one-half the size of the conventional measure.

Our estimates of the costs of taking liquidity are also upper bounds on the costs of trading by competing in the limit order book (LOB) because traders will only compete in the LOB if the expected costs of doing so, including the opportunity costs of unfilled limit orders, are less than the costs of taking liquidity. Investors who want to take liquidity by timing their trades do not need to wait long, as the option quote midpoint converges more than halfway to our estimate of option fair value in less than ten minutes.

Why do option market makers not update quotes frequently? Even if liquidity providers are faster than most liquidity takers, if they are slower than only one they are at risk to get “picked off.”⁵ To protect against this risk, market-makers post wider spreads that do not have to

⁵ An article in *Amsterdam Trader* describes an example of this in the market for the AEX (Netherlands stock index) options, writing (emphasis in original)

In the meantime, there’s something wrong in the AEX for a few weeks now. Several market makers noticed a steep increase in losing trades on their quotes. One market maker suggests a new competitor in the pit with faster systems and lower latency is able to arbitrage less sophisticated traders. “*In normal circumstances every liquidity provider was fast enough, now everyone is getting afraid to quote at all.*”

be changed with every change in the option fair value.⁶ Foucault, Roell, and Sandas (2003) model the tradeoff that dealers face between the cost of frequent quote revisions and the benefits of being picked off less frequently.⁷ It is also costly for option market makers to update quotes frequently because the options exchanges place caps on the number of quote updates and fine exchange members whose ratios of messages to executions is large. In addition, market frictions such as minimum tick sizes prevent market makers from continuously centering their quotes around the fair value. Finally, trades by execution timers incur a half-spread of about three cents, which exceeds market-makers' marginal costs of executing trades. Due to imperfect competition among liquidity providers, non-timers' trades are highly profitable for market makers, while the spreads on timers' trades appear to at least cover the marginal costs of trading. Thus, market makers can facilitate trading by cost sensitive investors by changing their quotes infrequently.

Finally, taking account of the reduction in trading costs due to execution timing can alter conclusions about the net of trading cost profitability of options trading strategies. We show this by backtesting two strategies that sell volatility by writing at-the-money straddles and holding them until expiration. The strategies are based on the difference between historical and implied volatility considered by Goyal and Saretto (2009) and the previous month stock return used by An, Ang, Bali, and Cakici (2014). Selling straddles according to these strategies produces high returns if transactions costs are disregarded. These strategies remain highly profitable, though of course less profitable, for traders who time executions and pay the algo effective half-spread. However, both strategies have statistically insignificant returns if one pays transactions costs equal to the conventional effective half-spread. These results provide guidance in interpreting the results of recent work on the returns of options trading strategies, including Driessen, Maenhout, and Vilkov (2009), Bali and Murray (2013), Cao and Han (2013), Doran, Fodor, and Jiang (2013), Boyer and Vorkink (2014), Muravyev (2016), and Cao, Han, Tong, and Zhan (2017), and also in assessing whether one can profitability trade on predicted option returns (Israelov and Kelly 2017). Our approach for estimating options fair values is also potentially useful in

⁶ Several models, including Biais, Foucault, and Moinas (2015), Foucault, Hombert, and Rosu (2016), and Jovanovic and Menkveld (2016), describe such behavior. Jones (2013) reviews these and other papers and points out that "the most common theme in these models is that HFT may increase adverse selection, which is bad for liquidity." He also writes "In most of these models, the downside of HFT is that their speed, or the information they collect and use for trading, increases adverse selection, thereby worsening liquidity."

⁷ Some execution timing can be interpreted as "pick-off" risk, which is discussed by Copeland and Galai (1983). See also Harris and Schultz (1997) and Battalio, Hatch, and Jennings (1997) who study trading by SOES bandits and pick-off risk in the equity market.

research that involves estimating and calibrating options pricing models to market prices (for example, Christoffersen, Fournier, and Jacobs 2017).

This paper is related to the literature that attempts to explain option bid-ask spreads using proxies for initial delta hedging costs, hedge rebalancing costs, and asymmetric information (Jameson and Wilhelm 1992, George and Longstaff 1993, Cho and Engle 1999, De Fontnouvelle et. al 2003, Kaul et al 2004, Petrella 2006, Engle and Neri 2010, and Goyenko et. al 2014). While this literature has had some success, our findings help explain why the success is less than complete because the theories are unlikely to explain the bias in the conventional effective spread that stems from using the midpoint as a proxy for the value.

1. Data and Summary Statistics

The main data are from Nanex, a firm specializing in delivering high quality data feeds. Nanex in turn obtains its data from standard sources: the Option Price Reporting Authority (OPRA) for options and the Securities Information Processor (SIP) for equities (TAQ data also are from the SIP). The Nanex data consist of all trades and intraday bid and ask quotes at one-minute frequency for all U.S. listed equity options and their underlying equities over the period from January 2004 to December 2015. The options quote data include the top of the limit order book for each options exchange, the stock quote data consists of the NBBO, and the timestamps are synchronized across markets. A limitation of this dataset is that in order to reduce storage requirements it includes only one minute snapshots of quotes rather than every quote update, and for each day includes only the quotes for options contracts with at least one trade during that day.⁸ Even so the dataset is very large: it is 15 TB in compressed form, and more than 100 TB uncompressed. This limits what we can do with the data. We merge the Nanex data with TAQ by stock ticker and date, which allows us to further merge it with CRSP using the TAQ-CRSP linkage provided by WRDS. We use TAQ data only to facilitate merging.

Table 1 reports summary statistics for the sample of trades in options on S&P 500 stocks during the sample period of January 2004 through December 2015. We report results for common stocks that are not in S&P 500 index in Internet Appendix Table IA.1. CRSP identifies S&P 500 constituents that we merge with the options data using the CRSP-TAQ linkage as described above. We include options trades with trade price above ten cents. Trades for which

⁸ Even with these limitations to reduce the size of the data, in uncompressed (compressed) form the data exceed 100 (15) terabytes.

trade direction is undetermined, the expected option price changes from one of the predictive models described below cannot be computed or are improbably large (have absolute value of more than 20% of the quote midpoint) are excluded. The first and last five minutes of trading are excluded to avoid the opening and closing rotations. We winsorize all variables at 0.1% to avoid the impact of outliers. After applying all filters, the final sample consists of approximately 621 million option trades, or an average of about 4.3 million trades per month.

The mean (median) trade price and size are 3.92 (1.72) dollars per share and 18.17 (6.26) contracts on one hundred shares each. The trade size distribution is highly skewed with 50th and 75th percentiles of 13.45 and 61.10 contracts respectively, while about 15% of trades have the smallest possible size of one contract. 39% of trades are in puts, and the typical option trade is somewhat out-of-the-money (OTM): the mean (median) of the absolute value of the option delta is 0.43 (0.42). However, many trades involve well in-the-money (ITM) options: 25%, 5%, and 1% of trades are in options with absolute value of delta greater than or equal to 0.57, 0.83, and 0.92, respectively. The distribution of time to expiration is highly skewed with a mean (median) of 82.30 (35.91) days. The mean (median) underlying stock price midpoint at the time of the trade and market capitalization are \$121.01 (\$62.59) and \$111,429 (\$58,978) million, respectively. There are more trades during the morning than the afternoon, as the median trade occurs 150 minutes after the beginning of trading.

The mean of the trade direction (buy = 1, sell = -1) is -0.034, which implies there are slightly more seller-initiated trades (51.7%) than buyer-initiated trades (48.3%). The trade direction is determined by the quote rule, and if a trade price is at the NBBO midpoint then the quote rule is applied to the best quotes of the reporting exchange. The mean (median) sizes quoted at the NBBO bid and ask are 1,202.9 (241.1) and 1,106.7 (217.2) contracts, much larger than the mean (median) trade size of 18.17 (6.26) contracts. On average, quotes on about 60% of the options exchanges match the NBBO at the time of a trade.

The last part of the table presents statistics summarizing the distribution of changes in the option midpoint over the ten minutes following a trade. The distribution of changes is approximately symmetric: for example, the median is zero, the 25th and 75th percentiles are -2.1% and 2.1%, and the 5th and 95th percentiles are -8.5% and 8.7%. The last two rows present information about the changes predicted by the models that are described below.

These statistics provide useful stylized facts about individual option trades. As in most markets, option trading activity is skewed, and a few stocks account for disproportionate number of trades. However, many applications of interest, including option trading strategies, require estimates of trading costs for the representative stock. Thus, we focus on a panel of 1,242,893 stock-days that represents averages across all option trades for given stock and day.

Table 2 presents estimates of the conventional measures of trading costs and price impact for the options on S&P 500 stocks. The statistics are the quoted half-spread at the time of the trade expressed in both percent (that is, the relative spread) and dollars, the conventional effective half-spread, and the percentage price impact. Panel A reports information about the measures for the full sample, while Panel B reports the same information for trades in OTM, ATM, and ITM options, where OTM options are those with absolute option delta $|\Delta| < 0.35$, ATM options are those with $0.35 \leq |\Delta| < 0.65$, and ITM options are those with $|\Delta| \geq 0.65$. We compute the statistics reported in the table as follows. We first compute the cost measures for each trade. Then, we use the option trades in each subsample (for example, OTM options) to compute the average for each stock-day. The table then reports the mean and other statistics describing the distribution of these stock-day measures. Thus, the means, for example, are equally weighted by stock-date, not by the number of trades.

The table reveals that conventional measures of transactions costs are large. For example, Panel A shows that in the full sample of options on S&P 500 stocks the mean and median of the stock-date quoted relative (dollar) half-spreads are 8.6% (13 cents) and 7.5% (10 cents), respectively. For some trades the spreads are much larger—the 95th percentile of the quoted relative (dollar) half-spread is 17.7% (32 cents). The effective spreads are about 80% of the size of the quoted spreads. The price impacts are also large: the mean (median) is 1.5% (1.3%), and the 95th percentile price impact is 3.8%. The OTM options in Panel B have higher relative spreads and lower dollar spreads, which is to be expected due to their lower prices. Perhaps unsurprisingly, the ATM options display spreads that are slightly smaller than those in the full sample (effective half-spreads of 5.3% and 9.7 cents). Then the ITM options have relative (dollar) spreads that are smaller (larger) than those of the options in the full sample, which also is to be expected due to the generally higher prices of these options. Dollar option spreads can sometimes be quite large. For example, the 95th percentile of the quoted (effective) half-spread

is 63 (41) cents. Internet Appendix Table IA.2 reports the corresponding results for options on non-S&P 500 stocks.

In addition to being large, conventional cost measures changed little over our sample period. Figure 1 is a time-series plot that shows the cross-sectional average conventional quoted and effective half-spreads over the sample period using all option trades on S&P 500 stocks. The quoted half-spread remains about 9% during most of the sample period, and increased to 11% toward the end of the period. Conventional effective half-spreads decreased from 8.4% to 5.8% by 2013 and then returned to 6.5%. The fact that the conventional measure decreased only slightly from 8.4% to 6.5% is puzzling as the options markets adopted technologies such as algorithmic trading that are known to have reduced transactions costs in the equity market and equity option volume tripled during the sample period according to the OCC. The figure also displays the time-series of adjusted effective half-spreads paid by algorithmic traders and the adjusted effective half-spread that averages over the spreads paid by both algorithmic and non-algorithmic traders. These spreads are described in Section 3.1.

The several panels in Internet Appendix Figure IA.2 display the time-series of dollar half-spreads for options on S&P 500 stocks, dollar and relative half-spreads of options on non-S&P 500 stocks, and relative half-spreads for OTM, ATM, and ITM options on S&P 500 stocks. The number of stocks in the non-S&P 500 sample varies between approximately 1,250 and 1,700 over the sample period.

2. Execution Timing in the Options Market

At high frequency changes in options prices are predictable. The key predictor is the difference between an estimate of the BSM option value, which is computed from an average of recent past implied volatilities and the current stock price, and the current quote midpoint. The idea is simple—when the BSM value exceeds (is less than) the current quote midpoint the option price is likely to increase (decrease). Lagged changes in stock and options prices and limit order book information improve the predictions. This section first explains how we compute the BSM value, and then shows that the difference between the BSM value and the current quote midpoint predicts short term changes in options quotes, and that the predictions can be improved by using the other covariates. Finally we show that investors exploit this predictability to time options trades, buying when the option fair value is close to the ask price and selling when the option fair value is close to the bid price.

2.1 Estimates of BSM options values

For each option and time t , we first use stock and option bid-ask quote midpoints at one-minute frequency over the previous 30 minutes and the BSM formula to compute 30 implied volatilities $IV_{t-i\Delta t}$, for $i = 1, 2, \dots, 30$ and $\Delta t = \text{one minute}$. We use the average of these 30 implied volatilities from times $t - i\Delta t$ as the estimate of the option volatility at time t :

$$\sigma_t = \frac{1}{30} \sum_{i=1}^{30} IV_{t-i\Delta t}.$$

We then use σ_t , the same BSM formula that was used to compute the implied volatilities, and the current stock price S_t to compute the BSM value at time t :

$$\hat{P}_t^{BSM}(S_t, K, T) = BSM_t(S_t, \sigma_t, K, T),$$

where $\hat{P}_t^{BSM}(S_t, K, T) = BSM_t(S_t, \sigma_t, K, T)$ is the option price computed using the BSM formula, K is the option strike price, and T is the time to expiration.⁹ The difference between the BSM value $\hat{P}_t^{BSM}(S_t, K, T)$ and the time t quote midpoint P_t is the key predictor.

While our approach uses the BSM formula, it does not rely on the assumption that the BSM formula is the correct option pricing model. Rather, it uses the formula for an option with strike K and time-to-expiration T to compute recent past implied volatilities of that option, and then uses the average of the past implied volatilities to compute the value of the same option for a possibly different stock price. Any non-normality in stock returns, options illiquidity (for example, Goyenko, Ornthanalai, and Tang (2014)), price pressure (Garleanu, Pedersen, and Poteshman (2008)), or other factor that impacts options prices will be subsumed into the implied volatilities, and then incorporated into the BSM value when the BSM formula is used to compute the BSM value.

It is intuitive that the difference between the BSM value and the quote midpoint predicts options price changes. For example, the BSM value will differ from the quote midpoint if the stock price has changed but options price quotes have not yet been updated or if a trader is temporarily quoting aggressively at either the bid or ask. In both cases the quote midpoint is likely to move toward the BSM value, in the first case because the options quotes are likely to be

⁹ The results depend little on the particular scheme used to compute the option volatility σ_t . As for the other parameters in the BSM formula, we assume no dividends and set the risk-free rate equal to 60-day LIBOR. Time to expiration is measured using calendar time. For a smaller subset of 38 stocks with available tick-by-tick options data, the results change little if we use a stock price with one second lag to allow for possible latency between the markets.

updated to follow the stock price and in the second case because any temporary aggressive quoting is likely to stop. On the other hand, any slowly changing variable that affects options market maker quotes, for example options illiquidity or market maker inventories, will enter into both the BSM value (because it impacts options quotes during the estimation window) and the quote midpoint at time t . Thus, slowly changing variables that have little impact on high frequency options price changes also do not impact the predictions, which is desirable.

Regardless, for execution timing what matters is whether the models predict options price changes, not why they do.

2.2 Short-term option price predictability

We first use a simple univariate regression

$$P_{t+\tau} - P_t = \alpha_0 + \alpha_1(\hat{P}_t^{BSM} - P_t) + \varepsilon_t \quad (2)$$

to show that the difference between the BSM value and the current quote midpoint predicts changes in options prices. Here $P_{t+\tau} - P_t$ is the change in the option mid-price over the time horizon τ and $\hat{P}_t^{BSM} - P_t$ is the difference between the BSM value and the time t option quote midpoint. We expect that $\alpha_1 > 0$, that is the option midpoint will move toward the BSM value.

We then extend the model to include other public information relevant for short-term option price dynamics: lagged delta-adjusted changes in the underlying stock price, lagged option price changes, and information from the limit order book. Specifically, we estimate the model:

$$P_{t+\tau} - P_t = \alpha_0 + \alpha_1(\hat{P}_t^{BSM} - P_t) + \alpha_2 \%ExchBid_t + \alpha_3 \%ExchAsk_t + \sum_{j=1}^2 \alpha_{j+3} \Delta \times (S_{t-(j-1)\Delta t} - S_{t-j\Delta t}) + \sum_{j=1}^2 \alpha_{j+5} (P_{t-(j-1)\Delta t} - P_{t-j\Delta t}) + \varepsilon_t, \quad (3)$$

where $\%ExchBid_t$ ($\%ExchAsk_t$) is the percentage of options exchanges with bid (ask) price equal to the National Best Bid (Offer), S_t is the underlying stock price midpoint at time t , $\Delta \times (S_{t-(j-1)\Delta t} - S_{t-j\Delta t})$ is the one-minute delta-adjusted change in the underlying stock price, and $P_{t-(j-1)\Delta t} - P_{t-j\Delta t}$ is the lagged change in the option quote midpoint. We expect that the option midpoint will move toward the BSM value ($\alpha_1 > 0$), follow liquidity supply and demand in the limit order book (LOB) ($\alpha_2 > 0$, $\alpha_3 < 0$), follow the delta-adjusted underlying stock price (α_5 , $\alpha_6 > 0$) as suggested by Muravyev et al. (2013), and revert after large option price changes (α_7 , $\alpha_8 < 0$). We refer to the models in equations (2) and (3) Models 1 and 2, respectively.

Below we use the forecast future price changes from the two models,

$$\hat{P}_{t+\tau} - P_t = \alpha_0 + \alpha_1(\hat{P}_t^{BSM} - P_t) \quad (4)$$

for Model 1 and

$$\begin{aligned} \hat{P}_{t+\tau} - P_t = & \alpha_0 + \alpha_1(\hat{P}_t^{BSM} - P_t) + \alpha_2 \%ExchBid_t + \alpha_3 \%ExchAsk_t \\ & + \sum_{j=1}^2 \alpha_{j+3} \Delta(S_{t-(j-1)\Delta t} - S_{t-j\Delta t}) + \sum_{j=1}^2 \alpha_{j+5} (P_{t-(j-1)\Delta t} - P_{t-j\Delta t}), \end{aligned} \quad (5)$$

for Model 2. Because the predicted price changes $\hat{P}_{t+\tau} - P_t$ use only information that is publicly available at time t we interpret the forecast $\hat{P}_{t+\tau}$, which equals the sum of the predicted change and the current quote midpoint, as an estimate of the option's fair value. This is intuitive—if the model predicts that an option price will change because stock prices have changed (and thus the BSM value has changed) but the options quotes have not yet been updated, then the predicted future midpoint is a better estimate of value than the current midpoint. Similarly, if the current midpoint is temporarily altered because a trader is quoting an aggressive bid or ask price and the model predicts that the midpoint will revert toward its previous level, then that predicted future midpoint is a better estimate of value than the current midpoint. By the Gauss–Markov theorem $\hat{P}_{t+\tau}$ is an unbiased estimate of the future midpoint even if $\hat{P}_t^{BSM}$ is biased, and thus our approach does not suffer from the limitations of the BSM model.

We estimate the two models for each option contract and trading day separately using quote data from regularly spaced one-minute intervals with the time horizon τ set to ten minutes. Thus, the model estimation step does not use any information about trades.¹⁰ We are fortunate to have sufficient computing resources to complete this computationally intensive step. We compute the average coefficients for each stock and day for the OTM, ATM, and ITM moneyness categories. Deltas are computed intraday with the BSM model and used to partition the options into the groups: OTM options are those with absolute option delta $|\Delta| < 0.35$, ATM options are those with $0.35 \leq |\Delta| < 0.65$, and ITM options are those with $|\Delta| \geq 0.65$. We then average the daily coefficient estimates for OTM, ATM, and ITM options across all stocks and months, and compute t -statistics for the average coefficient estimates using robust standard errors clustered by stock and month. Table 3 presents the average coefficient estimates, t -statistics, and regression R^2 s from the two models estimated using options on stocks in the S&P 500 index. The corresponding results for options on non-S&P 500 stocks are reported in Internet Appendix Table IA.3.

¹⁰ Few of the ten-minute intervals used in the model estimation contain trades (Internet Appendix Table IA.11).

The results for the univariate model (Model 1) show that the difference between the BSM value and the current midpoint $P_t^{\text{BSM}} - P_t$ predicts stock returns; the average coefficients on this variable are 0.36, 0.52, and 0.58 for OTM, ATM, and ITM options, and the t -statistics for tests of the hypothesis that the average slope coefficient equals zero are 67.6, 82.6, and 53.9, respectively. The predictability is economically important. For example, the average coefficient of 0.52 for ATM options implies that when the difference between the BSM value and the quote midpoint $P_t^{\text{BSM}} - P_t$ is 4 (–4) cents the midpoint is predicted to increase (decrease) by slightly more than two cents in the next ten minutes. The regression R^2 s of greater than seven percent indicate that the model explains an important fraction of the changes in option prices. The estimates of the intercepts, while statistically significant due to the large size of the data, are small in magnitude and not consistently either positive or negative.

The results for Model 2 in the right-hand part of Table 3 show that the difference $P_t^{\text{BSM}} - P_t$ continues to be an important predictor, though the estimated coefficients are somewhat smaller than those for Model 1. The delta-adjusted most recent stock price change $\Delta \times (S_t - S_{t-\Delta t})$ is also a highly significant predictor, with average coefficient estimates of 0.12, 0.13, and 0.088 for OTM, ATM, and ITM options and t -statistics of 68.2, 68.2, and 47.8. The average coefficient of 0.13 for ATM options indicates that a 5 cent increase in the stock price predicts that the price of an ATM option with $\Delta = 0.5$ will increase by $0.13 \times 0.5 \times 0.05 = 0.33$ cents. The coefficients on the second lag of the delta-adjusted stock price change are less than 40% the size of those on the first lag and predict correspondingly smaller option price changes. The percentages of exchanges at the National Best Bid and Offer (NBBO) are also economically important predictors. For example, the coefficients of 0.0250 and –0.0251 for ATM options imply that if 100% and 60% of exchanges are at best bid and offer, respectively, then the option price is predicted to increase by $100 \times 0.0250 - 60 \times 0.0251 \approx 1$ cent.

The average R^2 s for the different groups of options are between 12.9% and 16.9%, considerably larger than the averages of about 7% for Model 1. The predictability as measured by the R^2 s is remarkably large both in absolute terms and compared to the studies of equity market, which typically find R^2 s less than a few percent.¹¹ Of course the strongest predictor, the

¹¹ For example, Gao, Han, Li, and Zhou (2018) find an “impressive” R^2 of 1.6% in a regression of last half-hour return on first half-hour return for the SPY. Downing, Underwood, and Xing (2009) regress hourly stock returns on ten lags of stock and bond returns and find adjusted- R^2 s ranging from 0.0% for AAA-rated firms to 2.3% for BBB firms. Almgren, Thum, Hauptmann, and Li (2005) use information about a bank’s own orders and executions to predict price

difference between the BSM value and the quote midpoint $P_t^{\text{BSM}} - P_t$, has no analog in the equity market.

We next conduct extensive tests to ensure the predictability is robust and find that it is stable across stocks and over time. Panels A-C of Internet Appendix Figure IA.2 show the monthly average R^2 s for OTM, ATM, and ITM options from both models. The R^2 s for both models are relatively stable and commove together over the sample period. For ATM options, Model 1's R^2 varies from a maximum of 9.2% to a minimum of 5.6%. Model 2's R^2 varies from the maximum of 17.3% to the minimum of 10.1%. The results are similar for OTM options except the average R^2 is slightly higher. The R^2 for ITM options remains flat until it jumps from 13% to 17% in December 2013 for Model 2, with a smaller jump for Model 1. This jump coincides with the increase in the quoted spreads for these options in Figure 1. The R^2 for options on non-S&P 500 stocks in Panels D-F of Internet Appendix Figure IA.2 are also stable over time. Overall, option price predictability as measured by R^2 is persistent and remains large for every month in the sample.

Internet Appendix Table IA.9 examines the extent to which option price predictability as measured by R^2 varies with market conditions. We estimate stock-day panel regressions of R^2 on proxies for short and medium-term volatility, indicator variables for the earnings announcement day and the previous trading day, and stock fixed effects. The results for Models 1 and 2 are similar. Prices are less predictable when volatility is high and on earnings announcement and pre-announcement days, which are often associated with higher volatility. These results are consistent with the hypothesis that the variation explained by the predictive models is roughly constant, but unexplained variation increases with volatility leading to lower R^2 .

We also explore the out-of-sample performance of the predictive models in Table 4. To construct this table, we average the stock-by-day estimates of the model coefficients to compute estimates for each group (OTM, ATM, and ITM) of options for each month. Thus, for each month and model we have coefficient estimates for the OTM, ATM, and ITM options. Then, during the next month, for each option trade we use the pre-trade covariates and the estimates of the coefficients for the option's moneyness category from the previous month to predict the

changes during the period when orders are being executed and indicate that “The R^2 values for these [price impact] regressions are typically less than one percent.” Collins, Li, and Xie (2009) find that cross-sectional regressions in which earnings information is used to explain returns around earnings announcement yield single digit R^2 's (Table 4).

option price change over the ten-minute post-trade horizon. The results are robust to alternative ways of aggregating the historical coefficient values as the coefficients are persistent across stocks and time (Internet Appendix Figure IA.3.) Then for each moneyness category and month we pool the predictions and compute the usual out-of-sample R^2 defined as

$$R_{OS}^2 = 1 - \frac{\sum_{i=1}^N (dP_i - \widehat{dP}_i)^2}{\sum_{i=1}^N (dP_i - \overline{dP}_i)^2}, \quad (6)$$

where for the i th observation $dP_i = P_{t+\tau,i} - P_{t,i}$ is the actual change in the option price, $\widehat{dP}_i = \widehat{P}_{t+\tau,i} - P_{t,i}$ is the price change predicted by either Model 1 or 2, and $\overline{dP}_i$ is the predicted price change from a benchmark model. Campbell and Thompson (2007) use a similar measure to study the equity risk premium. Because our forecast horizon is only ten minutes, we use a simple benchmark of no change in the option price, that is we set $\overline{dP}_i = 0$ for each observation, which is also consistent with the summary statistics in Table 1. For each month we also compute another measure, the R^2 from a regression of the actual price changes dP_i on the forecasts $\widehat{dP}_i$ with the intercept suppressed. In these analyses, we do not condition on trade direction or use any other future information.¹²

Table 4 reports the means of the monthly R^2 s along with some information about their distribution. For Model 1, the mean out-of-sample R^2 is 5.5% and the median is 3.8%. The R^2 s from regressing the price changes on the forecasts are similar and the mean and median slope coefficients from these regressions are 0.86 and 0.90, reasonably close to one. For Model 2 the mean and median out-of-sample R^2 s are 7.0% and 5.4%, larger than those from Model 1. The R^2 s from regressing the price changes on the forecasts are larger, with mean and median R^2 s of 8.5% and 6.6%. Table IA.4 in the Appendix reports stronger results for non-S&P 500 stocks. There the average (median) out-of-sample R^2 s range from 9.7% (9.1%) for Model 1 to 12.3% (12.5%) for Model 2. The coefficients from regressing the price changes on the forecasts are 1.08 for Model 1 and 0.8 for Model 2. These results indicate significant out-of-sample predictability. Predictability typically decreases as the horizon shrinks, and at the daily horizon R^2 s of more than 2% or 3% are rare. Obtaining R^2 s greater than this at a ten-minute horizon indicates that the predictive models are successful.

¹² We would prefer to estimate out-of-sample R^2 s on one-minute quote snapshots instead of using the trades sample, but this is computationally difficult. Even the out-of-sample analysis using trades requires us to estimate regressions with an average of about 4.3 million option trade observations each month.

Lastly, Table IA.13 reports correlations between actual and predicted price changes. As expected, the forecasts of the two models are highly correlated with an average correlation of 70% for options on S&P 500 stocks, and 73% for non-S&P 500 stocks. The correlations between the actual price changes and the forecasts from Models 1 and 2 are 22% and 28%, respectively, for options on S&P 500 stocks, and 31% and 36% for options on non-S&P 500 stocks.

Overall, these analyses show that high-frequency intraday option price changes can be predicted using readily available public information. The predictability is consistent for both predictive models and is robust across moneyness categories and stocks, and over time. It can be used to compute the option fair value $\hat{P}_{t+\tau}$ at the current time t by combining current values of predictors, such as $(\hat{P}_t^{BSM} - P_t)$, with coefficients estimated on historical data.

2.3 Execution timing when option prices are predictable

If options prices tend to move toward the options' BSM values, then the difference between the BSM value and the current quote midpoint provides information about whether the current time is a good time to execute a trade. The additional covariates used in Model 2 improve the forecasts of price changes, and thus provide better estimates of whether now is a good time to trade. Investors who plan to buy options should execute purchases when the predicted future price $\hat{P}_{t+\tau}$ (that is, the estimate of option value) approaches the ask price. That is, they should buy when the difference between the transaction price they pay, the ask price, and the option value will be small. If they miss this opportunity, they will have to buy at a higher price as the midpoint is likely to increase. Similarly, investors who desire to sell options should execute sales when the predicted price $\hat{P}_{t+\tau}$ approaches the bid price, because doing so will make the difference between the value and the trade price they receive small. We refer to this opportunistic trade execution as execution timing.

Figure 2 is a stylized illustration of execution timing. The figure shows option prices (vertical axis) evolving in time (horizontal axis). The expected future midpoint (i.e., the option value) is shown in green. It evolves over time as the stock price, which is not shown, evolves. We assume that the current quote midpoint (grey) moves toward the option value (green) determined by the current price of the underlying. Execution timers will wait until the expected quote midpoint approaches the bid price (dark blue) to execute their sell trades, indicated by solid blue arrows. In this illustration investors timed their sales well because if they had waited

longer the bid price they receive would have decreased. The buy trade is not timed well because the mid-price is above the value at the time of the trade.

This example also illustrates why conventional measures of the bid-ask spread and price impact that use the current quote midpoint overestimate trading costs. Because the conventional effective spread uses the quote midpoint, it assumes that the sell trades in the figure incur the same costs as hypothetical buy trades executed at the same time. However, such buy trades are a poor trading decision because the investor could have waited for the expected decrease in price to take place and then buy at a better price. That is, the conventional measure of the effective spread fails to account for price predictability. More specifically, when the sell trades occur the current quote midpoint is above its expected future value, which is the estimate of option value. Using the higher current quote midpoint as a proxy for the option value overstates the effective spread and price impact of the trade. The quote midpoint is on average significantly higher than its expected value at the time of sell trades, and price will decrease even if no sell trades occur.

The appendix extends this example with a simple model that illustrates execution timing and how it affects trading costs. The model also shows that errors in estimating the option fair value are unlikely to spuriously create apparent execution timing or reductions in trading costs due to execution timing when execution timing does not exist.

Figure 3 shows that investors do actively engage in executing timing: they buy right before the price is expected to increase and sell before it is about to decrease. For this example, we use all 10,904 trades in options on Bank of America stock executed during January 2007. The blue line shows the predicted change in the option price (relative to the pre-trade mid-price) based on public information immediately before a trade computed using Model 2 (equation (5)). The predicted change is plotted as a function of the signed trade size, in option contracts. For positively signed trade sizes the option price is expected to increase by slightly more than 1% and increases in trade size to 1.5% for a 50-contract trade. Following negative signed trades, the expected option price change ranges from -0.6% for small trades to -1.2% for large trades. The grey line shows the option price changes during the ten minutes following a set of simulated trades for the same option and date at random times that do not overlap with the ten minutes periods following the time of an actual trade. As expected, the average option price changes following these random trades are close to zero. The black line shows the change in the option price midpoint from the time of a trade until ten minutes after, which is a popular measure of

price impact. It ranges in absolute value from 1.1% for small trades to 2% for large trades. The predicted price change conditional on a trade (the blue line) has the same sign as the actual change (the black line) but is smaller because option prices change due to inventory and adverse-selection impacts of option trades as discussed further in Section 3.2. Overall, execution timing (blue line) explains about two-thirds of the observed price impact in this example.

3. Execution Timing and the Cost of Taking Liquidity

Building on the above discussion, we can now measure the effect of execution timing on trading costs. The effective half spread, a widely used measure of the cost of taking liquidity, is the difference between the transaction price and the “true value” (Goettler, Parlour, and Rajan 2005) or “true underlying value of [the] security” (Bessembinder and Venkataraman 2010). In symbols it is $B_{t_i}(TP_{t_i} - V_{t_i})$, where TP_{t_i} is the trade price of trade i at time t_i , V_{t_i} is the fair value or “true value” at time t_i , and B_{t_i} is a buy/sell indicator that takes values $B_{t_i} = 1$ and $B_{t_i} = -1$ for buys and sells, respectively. Thus, computing the effective half spread requires an estimate of the fair value of the security, and empirical analysis requires “an observable proxy for the true underlying value of [the] security” (Bessembinder and Venkataraman 2010).

The bid-ask midpoint is often used as a proxy, leading to the conventional measure $B_{t_i}(TP_{t_i} - P_{t_i})$, where P_{t_i} is the bid-ask midpoint at time t_i . The average value of the conventional measure $B_{t_i}(TP_{t_i} - P_{t_i})$ will correctly estimate the average effective spread if $E(V_{t_i} - P_{t_i} | B_{t_i} = 1) = E(V_{t_i} - P_{t_i} | B_{t_i} = -1) = 0$, that is if the errors in using the midpoint P_{t_i} as a proxy for the value V_{t_i} are unrelated to the trade direction. For many markets it seems likely that, unconditionally, $E(V_{t_i} - P_{t_i}) = 0$, and the required conditions $E(V_{t_i} - P_{t_i} | B_{t_i} = 1) = E(V_{t_i} - P_{t_i} | B_{t_i} = -1) = 0$ may be satisfied in some markets. But in the options markets buys (sells) tend to occur when the fair value exceeds (is less than) the midpoint, and $E(V_{t_i} - P_{t_i} | B_{t_i} = 1) > 0$ and $E(V_{t_i} - P_{t_i} | B_{t_i} = -1) < 0$. To see this, recall that the predicted future price $\hat{P}_{t_i}$ from equations (4) and (5) for Models 1 and 2 is an estimate of the fair value, implying that the predicted future price change $\hat{P}_{t_i} - P_{t_i}$ is an estimate of $V_{t_i} - P_{t_i}$. Figure 3 shows that the average expected future price change $\hat{P}_{t_i} - P_{t_i}$ (blue line) is positive for buys and

negative for sells. This strongly suggests that for the options markets $E(V_{t_i} - P_{t_i} | B_{t_i} = 1) > 0$ and $E(V_{t_i} - P_{t_i} | B_{t_i} = -1) < 0$.¹³

Models 1 and 2 capture execution timing and provide the predicted future options prices $\hat{P}_{t_i}$ that are estimates of the fair value V_{t_i} . Thus, we introduce the adjusted effective half-spread $B_{t_i}(TP_{t_i} - \hat{P}_{t_i+\tau})$ as an alternative estimate of the cost of taking liquidity. Dividing by the current quote midpoint yields the relative adjusted effective half-spread for trade i executed at time t_i is

$$AdjES_{t_i} = B_{t_i}(TP_{t_i} - \hat{P}_{t_i+\tau})/P_{t_i}, \quad (7)$$

where we normalize by the midpoint so that our estimates are comparable to conventional effective spreads.

A simple example illustrates the difference between the conventional and adjusted effective spreads. Suppose that a call option is trading at 1.00/1.10 dollars bid/offer, and the quote midpoint is expected to increase by 1.5 cents in the next minute, from \$1.05 to \$1.065. An investor buys at the offer. The conventional measure of the effective half-spread for this trade is $1.10 - 1.05 = 5$ cents, while the actual cost as measured by the adjusted effective half-spread is only $1.10 - 1.065 = 3.5$ cents, less than the conventional measure. This example is typical—in our S&P 500 options data the mean difference between the conventional and adjusted effective half-spreads is 1.5 cents.

3.1 Cost of taking liquidity for trades that time executions

We now turn to estimating the effective spreads of liquidity-taking trades that do and do not reflect execution timing. For a trade i at time t , the difference between the predicted future price $\hat{P}_{t_i+\tau}$ and the current midpoint P_{t_i} , adjusted for trade direction B_{t_i} , is a measure of execution timing: $ET_i = (\hat{P}_{t_i+\tau} - P_{t_i})B_{t_i}$. Alternatively, dividing by the current midpoint $ET_i = (\hat{P}_{t_i+\tau} - P_{t_i})B_{t_i}/P_{t_i}$ measures the benefit of execution timing in relative terms. Holding other things constant, the larger the execution timing measure, the more likely the trade is executed by a trader who times executions.

¹³ Note that Models 1 and 2 do not condition on trades; they are estimated from the sample of quote snapshots, and do not use any trade information. The average future price change $\hat{P}_{t_i} - P_{t_i}$ is positive for buys and negative for sells because the publicly available covariates used in Models 1 and 2 are also correlated with trade direction, not because the models use information about trades.

For traders who do not time executions, the realized execution timing ET_i can be positive or negative, but we assume that its expectation is zero. This is definitional—trades that do not reflect timing are trades for which the expectation of the execution timing measure is zero. We also assume that for non-timed trades the expected price change is zero, which is reasonable because executions by non-timers are uncorrelated with changes in the option midpoint and options price changes are zero on average over short time intervals (Table 1). Because the average benefit of execution timing by non-timers is zero, the expected benefit of execution timing across both non-timers and timers can be estimated as the expectation across trades of the execution timing measure:

$$E(TP_{t_i} - P_{t_i})B_{t_i} - E(TP_{t_i} - \hat{P}_{t_i+\tau})B_{t_i} = E(\hat{P}_{t_i+\tau} - P_{t_i})B_{t_i}. \quad (8)$$

The left-hand side is the difference between the expectations of the conventional and adjusted effective half-spreads, that is the part of the conventional effective spread that can be saved by execution timing, and the right-hand side is the expectation timing measure. In terms of relative spreads,

$$E(TP_{t_i} - P_{t_i})B_{t_i}/P_{t_i} - E(TP_{t_i} - \hat{P}_{t_i+\tau})B_{t_i}/P_{t_i} = E(\hat{P}_{t_i+\tau} - P_{t_i})B_{t_i}/P_{t_i}. \quad (9)$$

The several expectations can be estimated by averaging over the sample trades indexed by i . This estimate of the cost savings from execution timing does not require any assumption about the distribution of execution timing, other than the assumption that for non-timed trades the expectation of the execution timing measure is zero.

To compute the share of trades that reflect execution timing we assume that for timed trades the timing measure is positive, that is $ET_i > 0$ if trade i is a timed trade. This also is definitional—any trade that does not have $ET_i > 0$ was not a timed trade. Thus, all trades for which $ET_i \leq 0$ come from traders who do not time executions. We use this, together with one additional assumption, to estimate the fraction of trades that reflect execution timing. Specifically, for trades that do not reflect any ability to predict high-frequency price changes, we assume that the median of the distribution of ET_i equals the mean, which is zero because $\hat{P}_{t+\tau,i}$ is the conditional expectation of P_i ; this implies that the median of the execution timing measure is also zero. This will be the case if the distribution of ET_i for non-timed trades is symmetric about zero, and can also be the case when the distribution is not symmetric. Given the approximate symmetry of high frequency option price changes reported in Table 1, the assumption that the

median of is zero for non-timed trades is reasonable. It implies that for each trade i with $ET_i \leq 0$ (that does not reflect execution timing) there is a “mirror” trade with $ET_i > 0$ that also does not reflect execution timing. Thus, in any period the number of trades that does not reflect execution timing ability is twice the number of trades with $ET_i \leq 0$ and the execution timing shares for trades that time execution can be estimated by

$$ETS = 1 - 2 \times \frac{1}{N} \sum_{i=1}^N I_{ET_i < 0} = 1 - 2E(I_{ET < 0}), \quad (10)$$

where ETS is the execution timing share, I_A is an indicator function of the set A , and N is the number of trades during the period. An advantage of this measure is that it can estimate the fraction of (liquidity-taking) algorithmic trades in a market using only public information.

An example illustrates the execution timing share computations. Assume the bid and ask prices are \$2.00 and \$2.10 with large quoted sizes, and that the current midquote of 2.05 equals the fair value. A small aggressive limit order to sell at 2.05 arrives. The new midquote is 2.025 but the fair value is still 2.05 because the midquote is expected to return to the previous level. Assume that no traders time executions, and that one buy and one sell trade arrive and are executed at the best bid and ask. After the executions, the bid remains at 2.00 and the ask returns to 2.10. The conventional effective half-spread is 2.5 cents for both trades: $|2 - 2.025|$ dollars for the sell, and $|2.05 - 2.025|$ dollars for the buy. The adjusted effective half-spread $AdjES$ is 5 cents $= |2 - 2.05|$ dollars for the sell and zero $= |2.05 - 2.05|$ for the buy, and the average is $2.5 = 0.5 \times 5 + 0.5 \times 0$ cents, the same as the conventional effective half-spread. The buy (sell) has a positive (negative) timing of 2.5 cents $= 2.05 - 2.025$ dollars. Thus, in this case with no traders who time executions, half of trades have positive realized timing and half have negative realized timing. The execution timing share is then $ETS = 1 - 2 \times 0.5 = 0$.

We compute the average adjusted effective half-spread by using the predictive models and equation (7) to compute the adjusted effective half-spread $AdjES_i$ for each trade i , and then averaging across trades. This averages across the spreads paid by timers and non-timers. Because they have no timing ability, the average effective half-spread paid by non-timers is simply the average conventional effective half-spread, denoted ES . To compute the effective half-spread paid by traders who time executions, we use the fact that the overall average

effective half-spread $AdjES$ can be written as an average of the half-spreads paid by timers ($AlgoES$) and non-timers (ES) weighted by their shares of trades:

$$AdjES = (1 - ETS) \times ES + ETS \times AlgoES \quad (11)$$

Rearranging this equation lets us estimate execution timers' cost of taking liquidity,

$$AlgoES = \left(\frac{1}{ETS} \right) (AdjES - (1 - ETS) \times ES). \quad (12)$$

One final point is that the benefits of execution timing depend on the quality of the predictive model. The estimates we compute below using equations (11) and (12) will underestimate the benefits of execution timing and overestimate the timers' effective half-spread $AlgoES$ and the overall adjusted effective half-spread $AdjES$ if the predictive model used by execution timers is superior to the predictive model we use to forecast the price change $\hat{P}_{t+\tau,i} - P_{t,i}$ and compute the estimates of $AdjES$ and $AlgoES$. Because it is likely that execution timers have access to more information than we use in our regression models, it is likely that that we underestimate the execution timing share ETS and overestimate the timers' and overall effective half-spreads $AlgoES$ and $AdjES$.¹⁴

We think it is unlikely that misspecification of our predictive models causes us to overestimate the execution timing share and underestimate $AlgoES$ and $AdjES$. The appendix shows that when there are no traders who time executions the trading cost measures do not depend on the estimate of fair value and thus do not depend on the predictive model. Because the trading cost measures do not depend on the predictive model when there are no execution timers, it is impossible for misspecification of the predictive model to create spurious apparent execution timing where there is none.

3.2 Results for the costs of taking liquidity

Table 5 Panel A presents the estimates of the costs of taking liquidity in options on S&P 500 stocks, while Panel B presents the estimates for OTM, ATM, and ITM options. For each estimate, we compute an average over all stocks and days where for each stock-day the estimate is an average over all option trades in that stock and moneyness category on that day. Internet Appendix Table IA.5 reports the corresponding results for non-S&P 500 stocks.

¹⁴ In addition, sophisticated traders with time-sensitive value-relevant information may sometimes take liquidity without using execution timing. Thus, the share of trades executed by sophisticated traders is likely greater than the share of trades that reflect execution timing.

The first row of results presents the mean and standard deviation of the relative quoted half-spread. In the full sample of all options the mean and median relative half-spread are 8.6% and 7.5% of the quote midpoint. Relative half-spreads of OTM options are larger due to the lower prices, while those of ATM and ITM options are smaller. The next set of results in each panel are for the conventional effective half-spreads. These are somewhat below the quoted half-spreads because trades sometimes occur within the quoted spread. We have already described conventional measures in Table 2, but Table 5 adds the spreads adjusted for execution timing. The next two rows report estimates of the adjusted effective half-spreads. The row labeled “Algo. adj. spread (%)” reports the estimates of the adjusted effective half-spread paid by the traders who time executions, while the row labeled “Adj. spread (%)” reports average effective half-spreads adjusted for execution timing as in equation (7) that reflect the average across the trades executed by both timers and non-timers. The next four rows present the corresponding dollar spreads. Finally, the last row provides an estimate of the fraction of trades that reflect execution timing.

Strikingly, estimates of execution timers’ costs of taking liquidity are much smaller than the corresponding conventional effective half-spreads. For example, in the sample of all options the algo adjusted effective half-spread is 2.5%, which is 37% of the conventionally measured effective half-spread of 6.8%. For a median stock-day, the difference is even more striking as the algo half-spread is 1.5%, or just a quarter of the conventional half-spread of 6%. In dollar terms, the average algo-adjusted and conventional half-spreads are 2.8 and 9.7 cents, respectively. The differences are larger in the samples of ATM and ITM options. Even in the set of OTM options the estimate of the algo spread is one-half the size of the conventional spread, 4.8% versus 9.6%. These indicate the conventional measures do a poor job of estimating costs of taking liquidity of sophisticated traders who time executions. Recognizing that the conventional effective half-spread is non-timers’ cost of taking liquidity, the results in Table 5 also make clear that timers and non-timers pay much different costs.

The adjusted effective half-spreads combine the costs of timers and non-timers and represent the overall average cost charged by liquidity providers. The adjusted effective half-spread of 5% is 26% lower than the conventional effective half-spread of 6.8%. Recalling that the non-timers pay the entire conventional effective half-spread, this lower spread is entirely due

to execution timers, who account for 38.2% of all trades and pay a 2.5% adjusted spread. In dollar terms, adjusted and conventional effective half-spreads are 6.8 and 9.7 cents, respectively. In aggregate, we estimate the cumulative benefits of execution timing using equation (8) to be about a billion dollars annually.

The execution timing that we document also helps explain the puzzling cross-sectional pattern of bid-ask spreads in which dollar spreads of ITM options are so much larger than those of OTM options. This is puzzling because if gamma and vega risks explain options bid-ask spreads, as is often suggested, then the dollar spread of an OTM call (put) should be equal to the dollar spread of the corresponding ITM put (call) because through the put-call parity relation the OTM call (put) and ITM put (call) have the same exposure to gamma and vega risks. Our results help resolve this puzzle as we find that the average difference between the dollar spreads of ITM and OTM options drops by a factor of more than two (from 7.8 to 3.5 cents) after accounting for execution timing as shown in Panel B of Table 5. Furthermore, trading costs for execution timers are similar (about three cents) for all moneyness categories.

Figure 1 shows that conventional effective spreads decreased only slightly, and quoted spreads even increased during our sample period despite improvements in technology that lead to significant cost reductions in the equity market and a tripling of equity option trading volume. However, the half-spreads for execution timers decreased from 3.8% to 1%, by almost a factor of four. In dollar terms, also half-spreads decreased from 3.6 cents to 1.3 cents, roughly matching the three times increase in options volume. Thus, the costs for sophisticated investors who can time executions decreased markedly, but the cost for non-timers such as retail investors stayed flat or sometimes even increased. Internet Appendix Figure IA.1 further documents trends in the conventional option trading cost measures.

After documenting the sample averages, we study how conventional and adjusted spreads vary across stocks based on several stock characteristics including market capitalization, volatility, quoted option bid-ask spread, and option call and put order imbalances. Specifically, Table 6 sorts all stocks (including non-S&P 500 stocks) on a given day into five portfolios with 253 stocks in each on average. We then compute the equally weighted averages of the spread measures for each portfolio and report the portfolio averages across days.

Overall, the main results depend little on the portfolio sorts and hold in every portfolio. In Panel A, stocks are sorted on market capitalization in the previous day, which varies from 429 to

42,935 million dollars. Unsurprisingly, bid-ask spreads are larger for small stocks. Quoted and conventional effective half-spreads are 14.8% and 11.3% for the small-capitalization portfolio and are 7.3% and 5.9% for large-capitalization stocks. Thus, the conventional spreads are large even for the largest stocks. The execution timing share is fairly uniform across market-capitalization quantiles at about 38%. Consistent with the pooled sample averages, execution timing reduces costs significantly. For large-cap stocks, the algo effective half-spread is 1.7% or about 28.8% of conventional effective half-spread of 5.9%. For small-capitalization stocks, the algo-adjusted effective half-spread is 7.3% or 64.6% of the conventional effective spread of 11.3%. Thus, small stocks are more expensive to trade due to higher fixed costs that affect costs for everyone including execution timers. The difference between the conventional and algo-adjusted half-spreads is about 5.5% in all quintiles.

Panel B sorts stocks on volatility, which is computed as the standard deviation of abnormal returns over the previous 21 trading days. Volatility does not affect conventional spreads much as an increase in the dollar spread due to higher volatility is offset by a higher option price. Conventional effective half-spreads increase from 8.0% to 9.9% for low to high volatility portfolios. The algo spread increases by about the same amount in absolute terms from 3.9% to 5.5%. Panel C sorts on option liquidity measured as the quoted half-spread, which varies from 21.1% for the low-liquidity portfolio to 4.7% for the high-liquidity quintile. Execution timing is much less frequent for the low-liquidity portfolio as only 25% of trades show execution timing. Adjusted half-spreads for execution timers are remarkably low for the high-liquidity portfolios, only 0.3% compared to conventional effective half-spreads of 4.0%. These results indicate that execution timing is important in all of the sorted portfolios.

Table 7 reports the results of regressions with stock fixed effects that show there is interesting time-series variation in execution timing. Spreads are regressed on indicators for the earnings announcement and pre-announcement days and short-term and mid-term volatility, which are measured as the previous day's absolute stock return and the standard deviation of abnormal returns over the last 21 trading days. As in the equity market, conventional option spreads are higher when volatility is high. However, algo adjusted spreads are actually lower when volatility is high. The effect is statistically significant, but the economic magnitudes are not large: if volatility doubles, conventional half-spreads increase by 0.2% and algo half-spreads decrease by 0.1%. This effect is stronger for non-S&P500 stocks. Perhaps higher volatility

makes it easier to time trades as deviations between the option value and the mid-price are larger and more frequent. The period around earnings announcements is associated with higher uncertainty. As a result, conventional half-spreads are wider by 0.40% on the announcement and pre-announcement days with t -statistics ranging from 11 to 14. However, the adjusted half-spread for timers is 1.15% lower than average, or 62% lower than its unconditional average of 2.5%. In untabulated results, we find that controlling for differences in option characteristics from day to day do not change these results. Execution timing share is slightly higher when volatility is high and slightly lower around earnings announcements.

4. Execution Timing and Price Impact

Prices respond to trades. The price impact of trade i is usually measured as the scaled difference $(V_{t+\tau} - V_t)/V_t$, where V_t and $V_{t+\tau}$ are the security's fair value or "true underlying value" as of the trade time t_i and at some time after the trade, $t + \tau$ (Bessembinder and Venkataraman 2009), and is intended to reflect "the market's assessment of the private information that the trade conveys" (Bessembinder 2003, p. 239). Empirical researchers often use the bid-ask midpoints at times t and $t + \tau$, P_t and $P_{t+\tau}$, as empirical proxies for V_t and $V_{t+\tau}$, leading to the use of $(P_{t+\tau} - P_t)/P_t$ to measure the price impact.

Table 8 shows estimates of this conventional measure of price impact using a ten-minute horizon for τ , which is common and gives the limit order book sufficient time to recover after a trade. For the sample of all options the quote midpoint moves on average by 1.5% of the option price in the first ten minutes after a trade. That is, a single option trade moves the price by a gigantic 1.5%. The median of 1.3% is smaller than the mean and the distribution of price impact is skewed to the right because sometimes trades are followed by large price movements. For OTM, ATM, and ITM options the means of the conventional measures are 1.9%, 1.29%, and 1.0% of the option prices, respectively. These price impacts are large in absolute terms, and larger than the estimates of price impact in the equity market. For example, Holden and Jacobsen (2014) show in the last column of their Table 1 Panel C that price impact for equity trades is 0.11%, or 14 times smaller than for options trades.

But if the price would have changed even absent the trade then only part of the price movement $(P_{t+\tau} - P_t)/P_t$ can be attributed to the trade and the conventional measure overstates the price impact. Equation (13) decomposes the observed price impact into two components, the

expected change in the quote midpoint that would have occurred even in the absence of a trade and the additional price impact that can be attributed to the trade,

$$P_{t+\tau} - P_t = (P_{t+\tau} - \hat{P}_t^{t+\tau}) + (\hat{P}_t^{t+\tau} - P_t), \quad (13)$$

where $\hat{P}_t^{t+\tau}$ is the expected value of $P_{t+\tau}$ conditional information available prior to the trade at time t . Thus, the second term on the right hand side $\hat{P}_t^{t+\tau} - P_t$ is an estimate of the price change that would have occurred even absent the trade at time t . Since $\hat{P}_t^{t+\tau}$ is the price that would have been observed even absent the trade, the first term $P_{t+\tau} - \hat{P}_t^{t+\tau}$ is the part of the price change that can be attributed to the trade and thus is a better measure of the price impact. We refer to this as the adjusted price impact because it has been adjusted to reflect the price movement that would have occurred even absent the trade at time t .

We compute estimates of this adjusted price impact measure $(P_{t+\tau} - \hat{P}_t^{t+\tau})/P_t$ using both Models 1 and 2. The results are reported in Table 8. Using Models 1 and 2 the mean price impacts in the full sample are 0.99% and 0.64% of the option price, approximately 65% and 42% as large as the means of the corresponding conventional measures. Table 8 also shows that the differences between the adjusted and conventional measures is large for ATM and ITM options. For example, for ITM options the adjusted price impact measures of 0.52% and 0.28% based on Models 1 and 2 are only 52% and 28% of the size of the conventional measure of 1.0%. Based on Model 2 that uses and limit order book information in addition to the Black-Scholes estimate of the option fair value the adjusted price impact measure is less than half the size of the conventional measure for all subsamples. These results show that reasonable estimates of the fair value computed using simple regression models result in estimates of price impact that are much smaller than the conventional measures.

We use portfolio sorts to study how the conventional and adjusted price impacts depend on stock and option characteristics. Table 9 reports the results when the adjusted price impacts are computed using Model 2. The first two columns report the mean conventional and adjusted price impacts for quintile portfolios sorted by the market capitalization. Both the conventional and adjusted measures of price impact are decreasing in market capitalization, with the adjusted price impact measure decreasing more strongly than the conventional measure. The mean adjusted price impact for the largest capitalization portfolio is only 0.5%, while the conventional price impact is 1.3% for this portfolio. The next two columns sort stocks by stock return volatility, from lowest (portfolio 1) to highest (portfolio 5). Unsurprisingly, underlying

stock volatility increases markedly from portfolio 1 to portfolio 5 (from 0.9% to 4.7% per day). The price impact measures increase in volatility but much less strongly, with the adjusted measure increasing from 0.9% of the option price in portfolio 1 to only 1.2% of the option price in portfolio 5. These results are consistent with the hypothesis that the information content of options trades is increasing in volatility, but that the relation between price impact and volatility is weak—the volatility of portfolio 5 is five times that of portfolio 1, but the adjusted price impact in portfolio 5 is only one-third larger than in portfolio 1. Next, the quintile portfolios are formed by sorting stocks by the average bid-ask spread of their options. The mean adjusted price impact increases considerably from the lowest to the highest spread portfolio, from 0.6% to 1.4% of the option price, or by a factor of greater than two. Still, it is much less sensitive to the quoted spread than the adjusted effected spread—from portfolio 1 to portfolio 5 the adjusted effective spread increases by a factor of 5.7, from 2.4% of the option price to 13.6%.

Similar to Table 7 for bid-ask spreads, Internet Appendix Table IA.14 reports how conventional and adjusted price impacts depend on volatility and earnings announcements using stock-date panel regressions with stock fixed effects. Both price impacts are higher when volatility is high and around earnings announcements.

5. Execution Timing and the After-Trading Cost Profitability of Options Trading Strategies.

In this section, we show how execution timing impacts the after-cost profitability of two trading strategies based on well-known option return predictors: the difference between historical and implied volatility ($HV - IV$) of Goyal and Saretto (2009) and the previous month's stock return used by An, Ang, Bali, and Cakici (2014).¹⁵ The idea behind the $HV - IV$ predictor is that if IV is high relative to HV options are overvalued and expected option returns are negative. The idea behind using the past underlying stock return as a predictor is that changes in volatility are negatively correlated with stock returns, but options prices do not fully reflect this correlation, that is the options market underreacts to the information in past stock returns. As a result, past underlying stock returns predict options returns. Strategies based on both predictors are profitable if one assumes that investors can trade at the quote midpoint, but are unprofitable if

¹⁵ An et. al (2014) do not directly study options returns. They show that the previous month's stock return predicts changes in call and put implied volatilities, which has implications for options prices and returns.

investors' trading costs are equal to the conventional effective bid-ask spread.¹⁶ We show that the reduction in transactions costs due to execution timing is large enough that the strategies are profitable for execution timers after taking account of their lower transactions costs.

As in the rest of the paper, we focus on S&P 500 index components during the period running from 2004 through 2015, which is the period for which we compute estimates of transactions costs. Options on most other (non-S&P 500) stocks are thinly traded, making them less interesting for most investors. That said, we provide results for non-S&P 500 stocks in Table IA.10 in the internet appendix.¹⁷

We use option prices from OptionMetrics and compute buy-and-hold options returns as in Goyal and Saretto (2009). Once a month, on the first trading day after the expiration date of the options on the regular expiration cycle, we consider the straddle that is closest to at-the-money and has approximately one month remaining to expiration. Specifically, the straddle consists of the call expiring in the next month on the regular expiration cycle with delta closest to 0.5 per share, together with the matching put. We hold this straddle through expiration and compute the before-trading cost option return as

$$OptRet_{t,t+T} = \frac{\max(S_{t+T} - K, 0) + \max(K - S_{t+T}, 0)}{c_t + p_t} - 1, \quad (14)$$

where c_t and p_t are the call and put mid-prices on the day the position is opened, S_{t+T} is the expiration date stock price, and $\max(S_{t+T} - K, 0)$ and $\max(K - S_{t+T}, 0)$ are the call and put option payoffs, respectively. We will also compute returns that incorporate estimates of the conventional and algo effective spreads, as described below. An advantage of computing buy-and-hold options returns as in Goyal and Saretto (2009) is that this avoids the transactions costs of closing the options position that would be incurred if we closed the position before expiration and costs of trading the stock that would arise if we used delta-hedge options returns. It also avoids the issues with options returns raised by Duarte, Jones, and Wang (2019).

¹⁶ We use $HV - IV$ and the previous month's return as predictors because they work during our sample period running from 2004 through 2015. Other potential options return predictors including idiosyncratic volatility, market capitalization, option effective spread, average option volume, stock beta, and stock turnover do not yield significant straddle returns during our sample period, even though some of these variables have been shown to predict option returns in the sample of all optionable stocks using data going back to 1996. One possible explanation is that options on S&P500 components are more efficiently priced than other options, especially in recent years.

¹⁷ Also, in untabulated results we replicated the Goyal and Saretto (2009) results for their sample and time period.

We apply mild filters and exclude options for which the bid price exceeds the ask price, the bid price is less than six cents, or there are special settlement conditions. Goyal and Saretto (2009) also require that the straddle is within 2.5% of at-the-money, specifically that the ratio of the strike to the underlying stock price is between 0.975 and 1.025. This filter is very restrictive as it reduces the sample size by a factor of 2.6, and for this reason we do not use it. In untabulated results we confirm that when we use this filter the option return magnitudes are similar to those reported in Table 10. We use the CRSP-OptionMetrics linkage provided by WRDS to merge option returns with our trading costs measures.

Because we select one straddle for each stock and month, the unit of observation is a stock-month. The final sample includes 56,100 stock-months for 742 unique stocks. On the purchase date, the median straddle has a delta of zero, a price of \$2.70, a strike price of \$45, 26 trading days remaining to expiration, an implied volatility of 31%, and an effective bid-ask half-spread of 5.8%. Ignoring transactions costs, the average straddle return is -2.3% .

For the *HV – IV* strategy we follow Goyal and Saretto (2009) and use the annualized standard deviation of daily stock returns over the previous 252 trading days as the measure of historical volatility, and compute *HV – IV* as the difference between this historical volatility and the 30-day at-the-money implied volatility from the OptionMetrics implied volatility surface. For the past stock returns strategy, we follow An et. al (2014) and use the stock return computed over the previous 21 trading days as the predictor.

We incorporate transactions costs by adjusting the returns using estimates of the conventional and algo effective half-spreads for trades in the options that comprise the straddle on the dates the positions are entered, the post-expiration Monday. If the data were sufficient, for each stock-month we would estimate the trading costs using trades in the options that comprise the straddle. But options trade infrequently, and there may be few (or no) trades in the options that comprise the straddle on the post-expiration Monday. Thus, we estimate trading costs using trades in all options with similar moneyness and maturity from the post-expiration Monday. Specifically, we use options with absolute delta between 0.35 and 0.65 that expire in 20 to 40 calendar days. For each straddle and date, the conventional and algo effective spreads are estimated as the averages of the spreads over all trades for these options on the given day. We also report average returns ignoring transactions costs by assuming that the trades in the options that comprise the straddle are executed at the quote midpoints.

Each month, we sort the stocks into ten portfolios using either $HV - IV$ or the previous month's stock return. Each portfolio includes straddles on about 40 stocks on average. The first row of Table 10 Panel A reports the equally-weighted average straddle returns of the portfolios sorted by $HV - IV$, ignoring transactions costs. The returns on some of these sorted portfolios are large; they range from -7.1% to -1.2% per month from the bottom to top decile, and the t -statistics below the average returns reveal that the average returns of portfolios 1 and 2 are significantly different from zero. The average return of the zero-cost portfolio formed by buying portfolio 10 and selling portfolio 1 is also large and significant, 5.9% per month with a t -statistic of 2.8. The last row of the panel reports the average value of the predictor $HV - IV$ in the decile portfolios.

The first row of Panel B reports the equally-weighted average returns of the portfolios sorted by the past underlying stock return. Similar to the results in Panel A, the returns to some of the portfolios are large. They range from -2.1% to -7.1% per month from the bottom to the top decile, and the t -statistics below the average returns reveal that the average returns of portfolios 9 and 10 are significantly different from zero. The average return of the zero-cost portfolio formed by buying portfolio 10 and selling portfolio 1 is also large and significant, -5.0% per month with a t -statistic of -2.5 .

These results that do not reflect transactions costs suggest that simple trading strategies based on either $HV - IV$ or past returns can be extremely profitable. Each month, one needs only sell the close-to-the-money straddles for which $HV - IV$ is low (portfolios 1 and 2 in Panel A) or for those which the past return is large (portfolios 9 and 10 in Panel B).

How do transaction costs affect these returns? In each panel, the third and fourth rows of results show that conventional and algo effective half-spreads average about 6% and 3% of the option prices, respectively, and do not vary much across the decile portfolios. Most of the decile portfolios for which results are reported in the top rows of Panels A and B show insignificant returns even before incorporating transactions costs, so these trading costs make them even less attractive. Thus, for the portfolios sorted by $HV - IV$ (past return) we focus on deciles 1 and 2 in Panel A (deciles 9 and 10 in Panel B) because these are the portfolios for which returns are large and statistically significant and might survive incorporating transactions costs. The returns to buying straddles are negative for the strategies based on both $HV - IV$ and past return, implying that investors should write straddles. Thus, we compute after-transaction cost returns by adding

an estimate of the half-spread to the option returns for each stock-month; this has the effect of reducing the returns to written straddles. Effective spreads are often used to evaluate after-cost performance of equity trading strategies (e.g., Lesmond, Schill, and Zhou (2003) and Korajczyk and Sadka (2004)) when only publicly intraday data such as TAQ are available.¹⁸ Because we compute the buy-and-hold return to holding straddles until expiration, we need only incorporate the initial transactions costs of entering the position.

The results in Panel A show that for the decile 1 portfolio sorted by $HV - IV$ incorporating transactions costs equal to the estimate of the conventional effective half-spread causes the pre-transaction cost return of selling straddles of 7.1% to decrease to an insignificant 1.3% (t -statistic = 0.7). That is, using the conventional effective spread the after-cost profits from this strategy are only 18% ($= 1.3/7.1$) as large as the pre-transactions cost returns and not statistically different from zero. The results for the decile 2 portfolio are even worse, as the reported average return of 1.4% per month implies that the returns from selling straddles are actually negative if a trader pays transactions costs equal to the conventional effective half-spread. Thus, this trading strategy is unattractive for investors who pay the conventional effective half-spread.

Execution timers pay lower trading costs. The next set of results labelled “Returns after algo eff. half-spread” captures this by using the algo effective half-spread to adjust the returns. Accounting for the algo effective half spread, returns of selling straddles in the decile 1 portfolio sorted by $HV - IV$ are of course worse than those that do not reflect transactions costs, but are still large at 4.0% per month and are statistically significant (t -statistic = 2.1). Based on the point estimate writing decile 2 straddles is also profitable, though the average decile 2 return is not statistically different from zero.

In Panel B, the after-transaction cost results for the decile 10 portfolio sorted using the past stock return are remarkably similar to those for the decile 1 portfolio in Panel A. The returns to the decile 9 portfolio in Panel B are somewhat better than the returns to the decile 2 portfolio in Panel A, but, like the decile 2 portfolio results in Panel A, are not statistically significant.

The results in Panels A and B indicate that execution timing can make a difference for the after-transaction cost profitability of options trading strategies. The returns from writing calls

¹⁸ Obviously, the bid-ask spread is only one component of transaction costs as price impact and commissions can also be important.

when $HV - IV$ is low (decile 1 in Panel A) or past return is high (decile 10 in Panel B) are high if one disregards transactions costs. These strategies remain highly profitable, though of course less profitable, for traders who can time executions and pay transactions costs equal to the algo effective half-spread. However, the portfolios have statistically insignificant returns if one pays transactions costs equal to the conventional effective half-spread. Put another way, transactions costs equal to the conventional effective half-spread consume about 80% of the very high pre-trading cost returns of the decile 1 (decile 10) portfolio in Panel A (B), but transactions costs equal to the algo effective half-spread consume only about 40% of the pre-trading cost returns of these portfolios.

6. Conclusion

We show that options prices are predictable at high frequency, and a large fraction of options traders exploit this predictability in timing their executions. The expected future quote midpoint computed from our best predictive model is a reasonable estimate of the option fair value. Traders who time executions when taking liquidity buy when this estimate of the fair value (the expected future midpoint) is close to but less than the quoted ask and sell when the estimate of the fair value is close to but greater than the quoted bid. Traders who exploit this predictability are able to take liquidity at low costs.

Measuring the cost of taking liquidity as the difference between the trade price and the estimate of option fair value at the time of the trade, adjusted for trade direction, the effective spread of the traders who time executions is on average 37.4% of the magnitude of the conventional estimate of the effective spread that uses the midpoint as a proxy for the fair value and 29.6% of the quoted spread. The overall average adjusted effective spread, averaging over the traders who do and do not time executions, is on average less than three-quarters as large as the conventional estimate and less than 60% of the quoted spread. These estimates of the costs of taking liquidity in the options markets help resolve the puzzle of why options trading volume is so high despite the seemingly high costs of taking liquidity—for traders who have access to execution algorithms that enable execution timing, the costs of taking liquidity are much lower than conventional estimates.

Taking account of the high frequency predictability of options prices also affects estimates of the price impact of options trades. Conventional measures of price impact that use the midpoint as a proxy for the fair value at large, on average about 1.5% of the option price for

options on S&P 500 stocks. Taking account of the predictability of options prices by using the predicted future option price as the estimate of the option fair value, the adjusted estimates of price impact are less than half as large as the conventional estimates.

Finally, taking account of the reduction in trading costs due to execution timing impacts conclusions about the net of trading cost profitability of options trading strategies. Two options strategies that sell volatility by writing at-the-money straddles and holding them until expiration produces high returns if transactions costs are disregarded, remain highly profitable, though of course less profitable, for traders who time executions and pay the effective half-spread of execution timers. However, these strategies statistically insignificant returns for a trader who pays the conventional effective half-spread.

Appendix

This appendix presents a simple model that generalizes the aggressive limit order example in the text. We first use the model to show that errors in estimating the option fair value (expected future midpoint) do not spuriously create apparent execution timing or reductions in trading costs due to execution timing when no traders time executions. We then illustrate the computation of the measures of timing and execution costs when some traders time executions.

A.1 Measures of execution timing and trading costs when no traders time executions

Let V denote the estimate of the option fair value (future midpoint) just prior to a trade, which might be either a buy or a sell, and let b , a , and $m = (b + a)/2$ denote the best bid, best offer, and the quote midpoint at the time of the trade. Assume that buys and sells are equally likely with probabilities $p = 1 - p = 0.5$. Use ET , ETS , $AEHS$, and $CEHS$ to denote execution timing, the execution timing share, the adjusted effective half-spread, and the conventional effective half-spread. For the trade, we have

$$ET = p(V - m) + (1 - p)(m - V) = 0.5(V - m) + 0.5(m - V) = 0, \quad (15)$$

$$ETS = 1 - 2 E(ET < 0) = 1 - 2 \times 0.5 = 0 \quad (16)$$

$$AEHS = p(a - V) + (1 - p)(V - b) = 0.5(a - V) + 0.5(V - b) = (a - b)/2, \quad (17)$$

$$CEHS = p(a - m) + (1 - p)(m - b) = (a - b)/2. \quad (18)$$

Half of trades have positive timing by chance, and the other half negative timing, also by chance, and thus the share of timed trades is zero as reflected in Equation (16). Note that the option midpoint does not need to match the option value even on average for the execution timing measure ET to be zero in this case.

Equations (17) and (18) assume that the quoted sizes at the best bid and offer are larger than the trade size so that the trade is executed at either the bid or ask. More generally, the buy (sell) quantity might exceed the national best offer (bid) quantity so that the buy (sell) is executed at multiple prices. For example, if the best ask is 5.20 and the next best offered price is

5.25 then part of a large buy order might be executed at 5.20 and part might be executed at 5.25.

Let $\alpha \geq a$ denote the average price at which a buy order is executed, and let $\beta \leq b$ denote the average price at which a sell order is executed. Then in this case ET and ETS are still given by equations (11) and (12) above while the formulas for $AEHS$ and $CEHS$ become

$$AEHS = p(\alpha - V) + (1 - p)(V - \beta) = 0.5(\alpha - V) + 0.5(V - \beta) = (\alpha - \beta)/2, \quad (19)$$

$$CEHS = p(\alpha - m) + (1 - p)(m - \beta) = 0.5(\alpha - m) + 0.5(m - \beta) = (\alpha - \beta)/2. \quad (20)$$

These differ from (17)-(18) above because the average execution prices can exceed (be less than) the ask (bid), that is $\alpha > a$ and $\beta < b$, with equality only if the probability that the trade size exceeds the best quoted sell or buy quantity is zero.

Equations (15), (16), (17), and (19) show that if there is no execution timing then ET , ETS , and $AEHS$ do not depend up the estimate of fair value V , and thus do not depend on any errors in the fair value or misspecification of the model used to estimate it. That is, misspecification of the regression model used to estimate the fair value cannot spuriously create an impression of execution timing when no traders time executions.

The assumption $p = 0.5$ is not required. With a bit of rewriting, the execution timing and $AEHS$ in equations (15) and (17) are

$$ET = (2p - 1)(V - m) \text{ and } AEHS = (1 - 2p)(V - m) + (a - b)/2, \quad (21)$$

For a single trade $ET = 0$ and $AEHS = CEHS$ if either $p = 0.5$ or $V = m$. Regardless of whether either or both of these conditions are satisfied, in expectation the two equations (21) become

$$E[ET] = (2p - 1)E(V - m) \text{ and } E[AEHS] = (1 - 2p)E(V - m) + E(a - b)/2. \quad (22)$$

These equations differ from the earlier equations because they include the terms $(2p - 1)E(V - m)$ and $(1 - 2p)E(V - m)$.

While $V = m$ is unlikely for any trade, if there is no execution timing it is likely that $E(V - m) = 0$, that is it is likely that on average the midpoint m equals the fair value V . Thus, if there is no execution timing the average measured execution timing in a large sample of trades should be zero even if $p \neq 0.5$. Further, note that even if $p \neq 0.5$ we expect $p \approx 0.5$, so that even if $2p - 1 \neq 0$ we expect $2p - 1 \approx 0$. Further, even if $E(V - m) \neq 0$ we expect $E(V - m) \approx 0$. Thus, the terms $(2p - 1)E(V - m)$ and $(1 - 2p)E(V - m)$ that appear in the equations above are the products of two small quantities, and even in the worst case should be close to zero. In this case the conventional and adjusted spreads $CEHS$ and $AEHS$ are close to each other, and thus Equation (8) in the paper implies that estimated execution timing should be close to zero too.

For the execution timing share, if $E(V - m) = 0$ and the median of $V - m$ is also zero then

$$ETS = 1 - 2 \times E(ET < 0) = 1 - 2(p \times 0.5 + (1 - p) \times 0.5) = 0. \quad (23)$$

A.2 Some traders time executions

Now we allow for execution timing. Assume that the next trade is by an execution timer with probability q , and by a non-timer with probability $1 - q$. The timer's trade is a buy if $V - m > 0$, and is a sell if $V - m < 0$, as timers only trade when the timing is positive for them. For a buy trade, the timing measure is $V - m$; for a sell trade it is $m - V = -(V - m)$. Thus, with probability q the timing measure is $|V - m|$ because execution timers always have non-negative timing.

The next trade is by a non-timer with probability $1 - q$, and the probabilities of buys and sells by non-timers are $p(1 - q)$ and $(1 - p)(1 - q)$. Conditional on the trade being by a non-timer, $E(V - m) = 0$. Considering these possible trades by timers and non-timers, the timing measure for the trade is

$$\begin{aligned} ET &= q|V - m| + (1 - q)p(V - m) + (1 - q)(1 - p)(m - V) \\ &= q|V - m| + (1 - q)(2p - 1)(V - m). \end{aligned} \quad (24)$$

Taking expectations, and using the fact that $E(V - m) = 0$ if the trade is by a non-timer, this equation simplifies to

$$E[ET] = qE(|V - m|),$$

the product of share of timers and their average timing benefit, which is of course positive.

The *AEHS* is

$$\begin{aligned} AEHS &= q((a - V)I_{V > m} + (V - b)I_{m \leq V}) + (1 - q)p(a - V) + (1 - q)(1 - p)(V - b) \\ &= q((a - b)/2 - |V - m|) + (1 - q)[(1 - 2p)(V - m) + (a - b)/2]. \end{aligned} \quad (25)$$

Taking expectations, and using the fact that $E(V - m) = 0$ if the trade is by a non-timer so that non-timers pay the spread $(a - b)/2$, the *AEHS* becomes

$$AEHS = -q|V - m| + (a - b)/2. \quad (26)$$

The *CEHS* is as usual

$$CEHS = (a - b)/2. \quad (27)$$

Thus, the adjusted spread differs from the conventional spread by the benefit of execution timing $q|V - m|$.

A.3 Execution timing share

The execution timing share is one minus twice the expected fraction of trades that display negative realized timing, that is

$$ETS = 1 - 2 \times E(ET < 0). \quad (28)$$

In estimation the expectation $E(ET < 0)$ is replaced by the fraction of shares that display negative timing. Conditional on the trade being by a non-timer, if the trade is a buy the timing measure is $V - m$ and if it is a sell the timing measure is $m - V$. Assume that for these non-timer buy trades the distribution of $V - m$ is symmetric about zero, and for the non-timer sell trades the distribution of $m - V$ is symmetric about zero. This implies that half of the trades by non-timers will display negative timing, that is $E(ET < 0) = (1 - q)/2$. Thus, the number of trades by non-timers is twice the number of trades with negative timing ability, that is $2(1 - q)/2$. Thus the fraction of trades by timers (the execution timing share) is

$$ETS = 1 - 2 \times E(ET < 0) = 1 - 2 \left(\frac{1 - q}{2} \right) = q. \quad (29)$$

The conclusion is that equation (28) correctly estimates share of trades by execution timers.

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Table 1**Summary statistics for options trades**

The sample consists of all trades in options on S&P 500 stocks from January 2004 through December 2015. Each statistic is computed for each sample month, and the table reports the average values of the statistics across the 144 sample months. Days-to-expiration is in calendar days. Trade price, trade size in number of contracts traded, and stock price are computed immediately prior to the option trade. Market capitalization of the underlying stock is as-of the previous day's close. Minute of day is the time at which the trade occurs, measured from 9:30 a.m. eastern time. Size quoted at ask and bid are the numbers of contracts quoted at the national best ask and bid immediately prior to the option trade. The percentages and numbers of options exchanges quoting at the ask and bid are also measured immediately prior to the trade. The sample contains a total of 620,916,282 options trades in S&P 500 stocks, or an average of 4,311,919 trades per month.

Variable	Mean	Std.dev.	Percentiles						
			1%	5%	25%	Median	75%	95%	99%
<u>Statistics of options trades</u>									
Call/put (call = 0, put = 1)	0.39	0.49	0	0	0	0	1	1	1
Buy/sell (buy = 1, sell = -1)	-0.034	1.00	-1	-1	-1	-1	1	1	1
Delta	0.43	0.21	0.07	0.12	0.27	0.42	0.57	0.83	0.92
Days-to-expiration	82.30	123.90	1.65	4.15	16.77	35.91	92.99	330.63	648.01
Trade price (\$)	3.92	7.03	0.16	0.27	0.78	1.72	4.10	14.64	32.95
Trade size (contracts)	18.17	70.46	1.00	1.00	2.01	6.26	13.45	61.10	214.98
Stock price (\$)	124.01	140.86	7.77	15.16	33.70	62.59	185.02	422.77	604.78
Market capitalization (\$million)	111,429	121,331	2,853	5,817	19,605	58,978	174,335	339,958	435,355
Minute of day (9:30 a.m. ET = 0)	169.1	118.0	4.60	12.15	60.57	150.78	276.02	364.27	377.71
<u>Market condition at time of trade</u>									
Size quoted at ask	1106.7	2,702.9	1.0	3.1	34.5	217.2	968.6	5,054.5	13,252.6
Size quoted at bid	1202.9	2,856.2	1.1	3.8	38.7	241.1	1,073.3	5,522.6	14,114.4
% of options exchanges at ask	58.2%	33.0%	13.0%	13.1%	23.5%	63.1%	90.3%	100.0%	100.0%
% of options exchanges at bid	59.1%	32.9%	13.0%	13.1%	24.6%	65.0%	91.0%	100.0%	100.0%
# of options exchanges at ask	4.49	2.67	1	1	1.69	4.71	6.97	8.22	8.28
# of options exchanges at bid	4.56	2.66	1	1	1.80	4.88	7.04	8.21	8.28
<u>Changes in options prices following trades</u>									
Actual change in quote midpoint	0.0%	5.1%	-14.8%	-8.5%	-2.1%	0.0%	2.1%	8.7%	15.2%
Change predicted by Model 1	0.0%	1.2%	-3.5%	-1.7%	-0.4%	0.0%	0.5%	1.9%	4.0%
Change predicted by Model 2	0.1%	2.0%	-5.8%	-3.0%	-0.7%	0.0%	0.9%	3.4%	6.3%

Table 2**Conventional measures of options trading costs**

The sample consists of all trades in options on S&P 500 stocks from January 2004 through December 2015. Each statistic is computed for each combination of underlying stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while Panel B uses only trades that fall in a particular moneyness category, which are defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The quoted half-spread is half the difference between the ask and bid prices at the time of the trade, expressed as either a percentage of the pre-trade bid-ask midpoint or in dollars. The effective half-spread is the difference between the trade price and the bid-ask midpoint (bid-ask midpoint and trade price) for trades signed as buys (sells), where a trade is signed as a buy (sell) if the trade price is greater (less) than the midpoint.

Panel A: All options on S&P 500 stocks

Bid-ask half-spreads	Mean	Std.dev.	Median
Quoted spread (%)	8.6%	4.9%	7.5%
Effective spread (%)	6.8%	3.9%	6.0%
Price impact (%)	1.5%	1.4%	1.3%
Quoted spread (\$)	0.13	0.14	0.10
Effective spread (\$)	0.10	0.09	0.08
Number of stock-days	1,242,893		

Panel B: Options split by moneyness

	<u>OTM</u>		<u>ATM</u>		<u>ITM</u>	
Bid-ask half-spreads	Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.
Quoted spread (%)	12.0%	6.5%	6.7%	4.6%	5.1%	4.0%
Effective spread (%)	9.6%	5.6%	5.3%	3.7%	3.7%	3.0%
Price impact (%)	1.9%	2.0%	1.3%	1.5%	1.0%	1.6%
Quoted spread (\$)	0.09	0.10	0.13	0.14	0.21	0.27
Effective spread (\$)	0.07	0.06	0.10	0.09	0.15	0.17
Num. of stock-days	1,110,085		1,174,130		1,068,654	

Table 3
Predictive model estimates

This table reports coefficient estimates and t -statistics for the predictive Models 1 and 2 in Equations (2) and (3) estimated using options on S&P 500 stocks. Model 1 predicts ten-minute changes in the option bid-ask midpoint $P_{t+\tau} - P_t$ using $\hat{P}_t^{BSM} - P_t$, the difference between the BSM value and the time t option quote midpoint. Model 2 also uses $\%ExchBid_t$ and $\%ExchAsk_t$, the percentages of options exchanges with bid or ask price equal to the National Best Bid (Offer), $\Delta \times (S_{t-(j-1)\Delta t} - S_{t-j\Delta t})$ for $j = 1$ and 2, the two most recent lagged one-minute delta-adjusted changes in the underlying stock price, and $P_{t-(j-1)\Delta t} - P_{t-j\Delta t}$, for $j = 1$ and 2, the two most recent lagged changes in the option quote midpoint. Moneyness categories are defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. We first estimate a separate regression for each option contract and day using snapshots of options quotes at one-minute frequency, without using any trade data. We then compute the average coefficient estimates for each stock and month and report the averages over the pooled stock-by-month panel. We list t -statistics based on robust standard errors clustered by stock and month in parentheses below the coefficient estimates. The sample consists of options on S&P 500 stocks during the period from January 2004 through December 2015.

	Model 1			Model 2		
	OTM	ATM	ITM	OTM	ATM	ITM
Constant	0.0003 (7.50)	-0.0001 (-2.50)	-0.0024 (-18.46)	-0.0003 (-2.73)	0.0001 (0.00)	-0.0043 (-13.03)
$\hat{P}_t^{BSM} - P_t$	0.3637 (67.60)	0.5155 (82.61)	0.5789 (53.90)	0.2427 (56.44)	0.3697 (68.46)	0.4673 (49.09)
$\%ExchBid_t$				0.0174 (47.03)	0.025 (35.21)	0.0385 (21.75)
$\%ExchAsk_t$				-0.0167 (-40.73)	-0.0251 (-32.60)	-0.0356 (-21.98)
$\Delta \times (S_t - S_{t-\Delta t})$				0.1228 (68.22)	0.1329 (68.15)	0.088 (47.83)
$\Delta \times (S_{t-\Delta t} - S_{t-2\Delta t})$				0.0482 (30.31)	0.0494 (26.14)	0.0265 (15.14)
$P_t - P_{t-\Delta t}$				-0.0456 (-16.76)	-0.0544 (-16.39)	-0.0205 (-6.19)
$P_{t-\Delta t} - P_{t-2\Delta t}$				-0.0124 (-6.05)	-0.0178 (-6.38)	-0.0027 (-0.90)
Adj R^2	0.080	0.075	0.076	0.169	0.134	0.129
N	70,809	71,242	71,376	70,018	70,524	70,614

Table 4**Predictive model out-of-sample performance**

This table reports measures of the out-of-sample performance of the predictive models. The results for the out-of-sample R^2 s in the first row of results are based the formula for the out-of-sample R^2 in Equation (4). For each sample month, we estimate the R^2 s using all option trades on S&P 500 stocks during the month and then report the mean, median, and standard deviation of the monthly R^2 s. The last three rows report results from no-intercept regressions of changes in mid-quote prices on the ex-ante forecasts from each model. We estimate regression separately for each month and report the statistics for the time series of monthly estimates. In both cases, the Model 1 and 2 forecasts are computed using the pre-trade values of the covariates and the coefficient estimates from Equations (2) and (3) estimated using data from the previous month. The sample consists of options on S&P 500 stocks during the period from January 2004 through December 2015.

Parameter	Model 1			Model 2		
	Mean	Std.dev.	Median	Mean	Std.dev.	Median
Out-of-sample R^2	5.5%	4.4%	3.8%	7.0%	6.1%	5.4%
<u>Regression of realized price change on forecast</u>						
b	0.86	0.18	0.90	0.69	0.13	0.69
t -statistic	(401.65)	(88.49)	(397.57)	(533.39)	(87.53)	(529.97)
R^2	5.6%	4.2%	3.7%	8.5%	5.2%	6.6%
No. obs. each month	4,309,000	2,162,572	4,570,654	4,287,835	2,147,651	4,563,637

Table 5
Trading cost estimates

This table reports various estimates of options trading costs and the share of trades that reflect execution timing. Each statistic is computed for each combination of stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while Panel B are for trades in the given moneyness category defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The quoted half-spread is half the difference between the ask and bid prices at the time of the trade, expressed as either a percentage of the pre-trade bid-ask midpoint or in dollars. The effective half-spread is the difference between the trade price and the bid-ask midpoint (bid-ask midpoint and trade price) for trades signed as buys (sells). The adjusted effective half-spread is computed using Equation (7) and represents average costs across execution timers and non-timers. The cost for execution timers (the algo half-spread) is computed using Equation (12). The execution timing share is the fraction of trades that show timing and is computed using Equation (6). The sample consists of all trades in options on S&P 500 stocks from January 2004 through December 2015.

Panel A: All options

Bid-ask half-spreads	Mean	Std.dev.	Median
Quoted half-spread (%)	8.6%	4.9%	7.5%
Effective half-spread (%)	6.8%	3.9%	6.0%
Adjusted half-spread (%)	5.0%	4.0%	4.0%
Algo. half-spread (%)	2.5%	5.7%	1.5%
Quoted half-spread (\$)	0.129	0.138	0.095
Effective half-spread (\$)	0.097	0.089	0.076
Adjusted half-spread (\$)	0.068	0.078	0.049
Algo. half-spread (\$)	0.028	0.096	0.015
Execution timing share (%)	38.2%	30.3%	39.8%
Number of stock-days	1,242,893		

Panel B: Averages by moneyness category

Bid-ask half-spreads	OTM	ATM	ITM
Quoted half-spread (%)	12.0%	6.7%	5.1%
Effective half-spread (%)	9.6%	5.3%	3.7%
Adjusted half-spread (%)	7.6%	3.6%	2.3%
Algo. half-spread (%)	4.8%	1.5%	0.7%
Quoted half-spread (\$)	0.093	0.128	0.211
Effective half-spread (\$)	0.071	0.097	0.149
Adjusted half-spread (\$)	0.056	0.066	0.091
Algo. half-spread (\$)	0.034	0.026	0.028
Execution timing share (%)	38.1%	39.6%	42.2%
Number of stock-days	1,110,085	1,174,130	1,068,654

Table 6**Options trading costs for various groups of stocks**

This table reports average options trading costs in portfolios of stocks formed by sorting stocks on various characteristics. Each day, we sort stocks into five portfolios based on either market capitalization on the previous day in Panel A, stock volatility estimated as the standard deviation of abnormal returns over the previous 21 days in Panel B, and quoted option half-spread in Panel C. The table reports the quoted half-spread, the conventional effective half-spreads, the adjusted effective half-spread, and the algo effective half-spread. We first compute the portfolio averages for each day and then report the overall averages across days. The last two columns report the fraction of trades with execution timing and the average values of the variable used to sort the stocks into portfolios. The sample consists of options on all common stocks (not just S&P 500 stocks) during the period from January 2004 through December 2015, and thus each portfolio has slightly more than 253 stocks. The last row of each panel reports the difference between portfolios 5 and 1.

Panel A: Portfolios sorted on underlying stock market capitalization

<u>Trading cost measures (half-spreads)</u>						Market capitalization (\$million)
Portfolio	Quoted spread	Effective spread	Adj. eff. spread	Algo. spread	Exec. timing %	
1	0.148	0.113	0.093	0.073	34%	428.791
2	0.128	0.097	0.077	0.054	37%	1,283.204
3	0.117	0.089	0.068	0.045	38%	2,733.532
4	0.101	0.078	0.059	0.034	39%	6,953.269
5	0.073	0.059	0.042	0.017	38%	42,934.560
Difference	-0.0746	-0.0543	-0.0511	-0.0556	3.56%	42505.773

Panel B: Portfolios sorted on underlying stock volatility

<u>Trading cost measures (half-spreads)</u>						Volatility
Portfolio	Quoted spread	Effective spread	Adj. eff. spread	Algo spread	Exec. timing %	
1	0.103	0.080	0.062	0.039	38%	0.009
2	0.107	0.082	0.063	0.040	38%	0.014
3	0.111	0.085	0.065	0.042	38%	0.018
4	0.118	0.090	0.070	0.046	37%	0.025
5	0.128	0.099	0.079	0.055	35%	0.044
Difference	0.0252	0.019	0.0177	0.0156	-2.52%	0.035

Panel C: Portfolios sorted on options quoted spread

<u>Trading cost measures (half-spreads)</u>					
Portfolio	Quoted spread	Effective spread	Adj. eff. spread	Algo spread	Exec. timing %
1	0.047	0.040	0.024	0.003	40%
2	0.076	0.061	0.041	0.016	42%
3	0.101	0.079	0.058	0.032	40%
4	0.132	0.101	0.081	0.056	36%
5	0.211	0.155	0.136	0.117	27%
Difference	0.1639	0.1151	0.1126	0.1138	-12.8%

Table 7**Time-series variation in options trading costs**

This table presents regression results showing how options trading costs respond to time-series shocks to stock volatility and information asymmetry. We estimate stock-day panel regression with stock fixed effects and dependent variables consisting of relative conventional and algo effective half-spreads, their difference, and the execution timing share. The independent variables include short and mid-term volatility measures: the absolute abnormal stock return from the previous day ($|StkRet_{t-1}|$) and the standard deviation of abnormal stock returns in the previous 21 days ($Std(AR)_{1m}$), and measures of asymmetric information: indicator variables for the earnings and pre-earnings announcement days (EAD_t, EAD_{t-1}). The sample consists of options on S&P 500 stocks during the period from January 2004 through December 2015. The panel includes 1,242,874 stock-days and standard errors are clustered by stock and by day.

	Conv. half- spread	Algo half- spread	Difference	Exec. timing share
$ StkRet_{t-1} $	0.0141 [2.3]	-0.0657 [-9.3]	0.0798 [18.6]	0.2652 [8.1]
$Std(AR)_{1m}$	0.1285 [6.7]	-0.0124 [-0.6]	0.1409 [12.4]	1.2704 [10.8]
EAD_t	0.004 [14.5]	-0.0115 [-23.7]	0.0155 [34.5]	-0.022 [-7.9]
EAD_{t-1}	0.004 [11.2]	-0.0015 [-3.4]	0.0055 [14.9]	-0.0293 [-11.0]
R^2	33%	15%	2%	3%

Table 8**Estimates of price impact**

Conventional and adjusted measures of price impact. The price impact of a buy trade is the difference between the midpoint ten minutes after the trade and the pre-trade midpoint, expressed as a percentage of the pre-trade midpoint, or the negative of this for a sell trade. Adjusted price impacts are computed using Equation (9) by subtracting the expected change in the option price computed using Model 1 or 2 from the actual price change during the ten minute period following a trade. Each statistic is computed for each combination of stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while each set of results in Panel B uses only trades for the indicated moneyness category, which is defined by absolute option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The sample consists of all trades in options on S&P 500 stocks during the period from January 2004 through December 2015.

Panel A: All options

Price impact	Mean	Std.dev.	Median
Conventional measure	1.526%	1.363%	1.308%
Adjusted using Model 1	0.989%	1.222%	0.790%
Adjusted using Model 2	0.644%	1.189%	0.467%
No. of stock-days	1,242,893		

Panel B: By moneyness category

Price impact	OTM	ATM	ITM
Conventional measure	1.945%	1.286%	1.004%
Adjusted using Model 1	1.370%	0.764%	0.519%
Adjusted using Model 2	0.919%	0.463%	0.283%
No. of stock-days	1,110,085	1,174,130	1,068,654

Table 9**Price impact for various groups of stocks**

This table reports estimates of price impact in portfolios of stocks formed by sorting stocks on various characteristics. We sort stocks into five portfolios based on either market capitalization on the previous day in Panel A, stock volatility estimated as the standard deviation of abnormal returns over the last 21 days in Panel B, and quoted option half-spread in Panel C. The table reports estimates of the conventional price impact and the adjusted price impact adjusted with the adjustment done using Model 2. We first compute the portfolio averages for each day and then report the overall averages across days. The sample consists of options on all common stocks (not just S&P 500 stocks) during the period from January 2004 through December 2015, and thus each portfolio has slightly more than 253 stocks. The last row reports the difference between portfolios 5 and 1 and the corresponding *t*-statistic adjusted for heteroscedasticity and autocorrelation. All averages reported in the table are statistically significant due to the large sample sizes.

Sorting variable:	Market capitalization		Volatility		Quoted option spread	
	<u>Price impact</u>		<u>Price impact</u>		<u>Price impact</u>	
Portfolio	Conventional	Adjusted	Conventional	Adjusted	Conventional	Adjusted
1 (Low)	0.025	0.015	0.018	0.009	0.014	0.006
2	0.022	0.012	0.019	0.009	0.018	0.008
3	0.020	0.010	0.020	0.010	0.021	0.010
4	0.018	0.008	0.021	0.011	0.023	0.013
5 (High)	0.013	0.005	0.022	0.012	0.023	0.014
High – Low	-0.0115	-0.0103	0.0043	0.0038	0.0083	0.0079
	[-118.8]	[-185.6]	[48.6]	[79.5]	[77.0]	[121.1]

Table 10**Impact of execution timing on the after-cost profitability of options trading strategies**

Returns from selling straddles with several assumptions about transactions costs. On the trading day following the expiration of options on the regular cycle, we buy a straddle consisting of a call and a put with the same strike and expiration in about one month and hold the straddle until it expires. We compute straddle returns following Equation (10). We form ten portfolios each month and report portfolio average return over all months. Panels A and B report the equally-weighted average returns of portfolios sorted by the difference between historical and implied volatility ($HV - IV$) and by the previous month's stock return, respectively. The first row of results reports straddle returns are assuming that trades are at mid-quote prices, and thus assumes that there are no transactions costs. The next two rows adjust returns using estimates of conventional and algo. effective half-spreads by adding the half-spreads to the no transactions costs straddle returns. The last three rows of the table report the estimates of the transactions costs and the average values of the sorting variable ($HV - IV$ or the previous month's stock return) for the portfolios. The t -statistics are based on standard errors adjusted for heteroscedasticity and autocorrelation. The sample period runs from January 2004 through December 2015.

Panel A: Portfolios sorted on $HV - IV$

Portfolio:	L= 1	2	3	4	5	6	7	8	9	H = 10	H - L
<u>Straddle returns:</u>											
No transactions costs	-0.071***	-0.046**	-0.032	-0.022	-0.011	-0.024	0.003	-0.028	-0.026	-0.012	0.059***
	[-3.7]	[-2.1]	[-1.3]	[-0.9]	[-0.4]	[-0.9]	[0.1]	[-1.0]	[-0.9]	[-0.4]	[2.8]
Algo eff. half-spread	-0.040**	-0.014	0								
	[-2.1]	[-0.6]	[0.0]								
Conven. eff. half-spread	-0.013	0.014	0.028								
	[-0.7]	[0.6]	[1.2]								
<u>Other statistics:</u>											
Conv, eff. half-spread	0.059	0.06	0.06	0.06	0.059	0.06	0.06	0.059	0.058	0.06	0.001
Algo eff. half-spread	0.031	0.032	0.033	0.032	0.031	0.032	0.031	0.03	0.03	0.031	0
$HV - IV$	-0.108	-0.046	-0.026	-0.013	-0.001	0.011	0.024	0.04	0.063	0.133	0.241

Panel B: Portfolios sorted on previous month's stock return

Portfolio:	L= 1	2	3	4	5	6	7	8	9	H = 10	H – L
Straddle returns:											
No transactions costs	–0.021	–0.007	–0.005	–0.025	–0.014	–0.008	–0.027	–0.03	–0.061***	–0.071***	–0.050**
	[–0.8]	[–0.3]	[–0.2]	[–1.0]	[–0.6]	[–0.3]	[–1.1]	[–1.2]	[–2.6]	[–3.2]	[–2.5]
Algo eff. half-spread								0.001	–0.03	–0.044**	–0.052**
								[0.0]	[–1.3]	[–2.0]	
Conven. eff. half-spread								0.029	–0.003	–0.015	–0.053***
								[1.1]	[–0.1]	[–0.7]	
Other statistics:											
Conv, eff. half-spread	0.059	0.061	0.061	0.061	0.06	0.061	0.059	0.059	0.058	0.056	–0.003
Algo eff. half-spread	0.03	0.032	0.034	0.033	0.032	0.033	0.03	0.031	0.031	0.028	–0.002
Previous month's stock return	–0.135	–0.062	–0.035	–0.016	0	0.015	0.03	0.048	0.073	0.137	0.272

Figure 1

Options trading costs over time

This figure shows the evolution of the average quoted half-spread (light grey), the conventional effective half-spread (grey), the adjusted effective half-spread (dark grey), and the algo effective half-spread (black) over the period from January 2004 to December 2015. The sample consists of options on S&P 500 stocks during the period from January 2004 through December 2015. All of the half-spreads are normalized by the pre-trade midpoint price. Each point on the graph is a 21-day moving average of the averages across all option trades on a given day. The quoted half-spread is half the difference between the ask and bid prices at the time of the trade, expressed as a percentage of the pre-trade bid-ask midpoint. The effective half-spread is the difference between the trade price and the bid-ask midpoint (bid-ask midpoint and trade price) for trades signed as buys (sells). The adjusted effective half-spread is computed using Equation (7) and represents average costs across execution timers and non-timers. The also effective half-spread is computed using Equation (12).

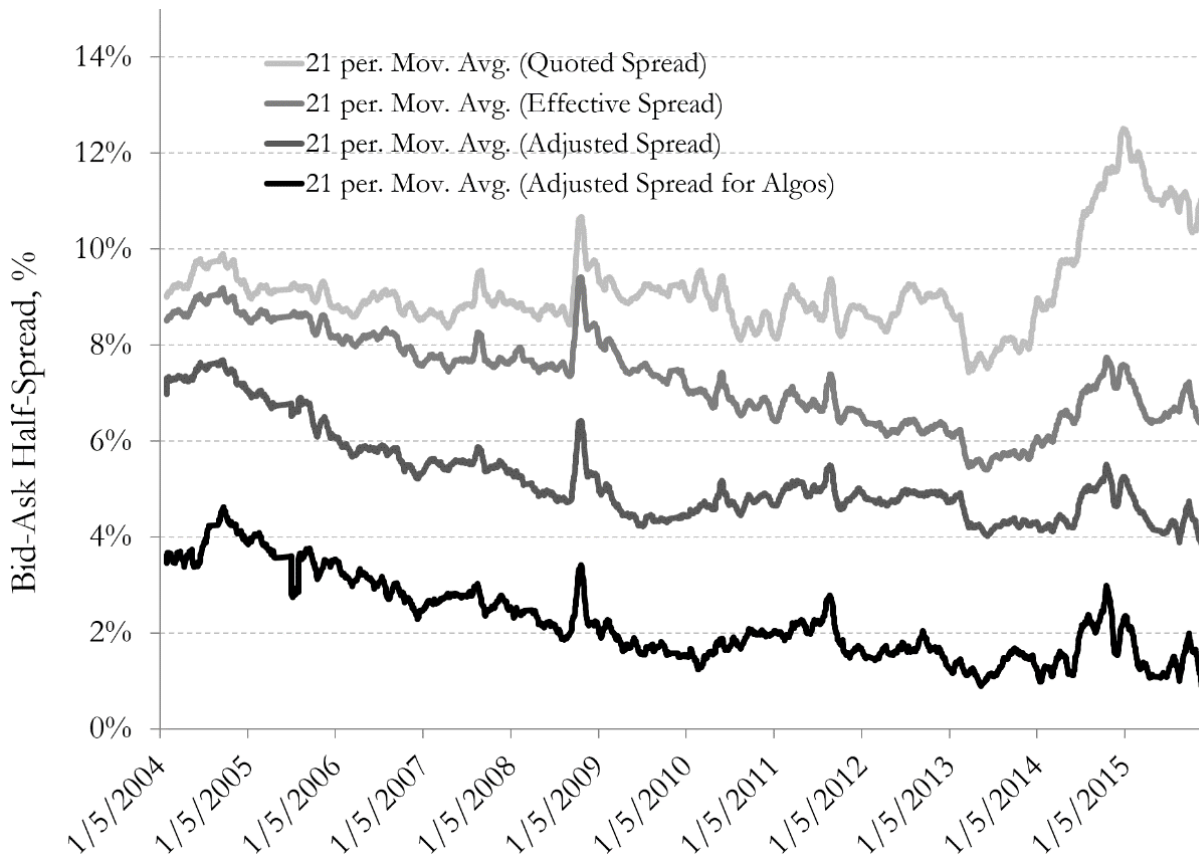


Figure 2

Illustration of execution timing

Option prices (vertical axis) evolve over time (horizontal axis). The current midpoint price (dashed grey) eventually converges to the future expected midpoint (green) implied by the current underlying price, which is the option fair value from the predictive model in Equation (2). Arrows denote time and direction of option trades that occur at the bid price or the ask price. The sellers waited until the expected quote midpoint approached the bid price (dark blue) to execute their trades. The two sales are timed well because had the sellers waited, the bid price would have decreased increasing their costs. The buy trade is not timed well as option value is below the midpoint price at the trade time. Conventional bid-ask spread measures rely on the midpoint price and thus are the same for buys and sell trades in this example.

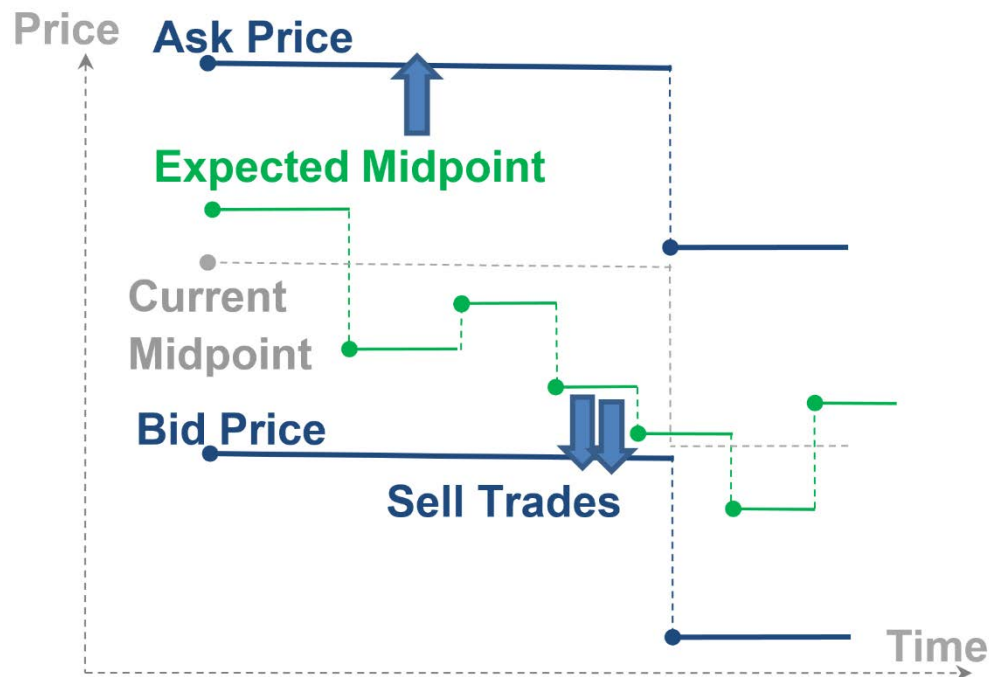
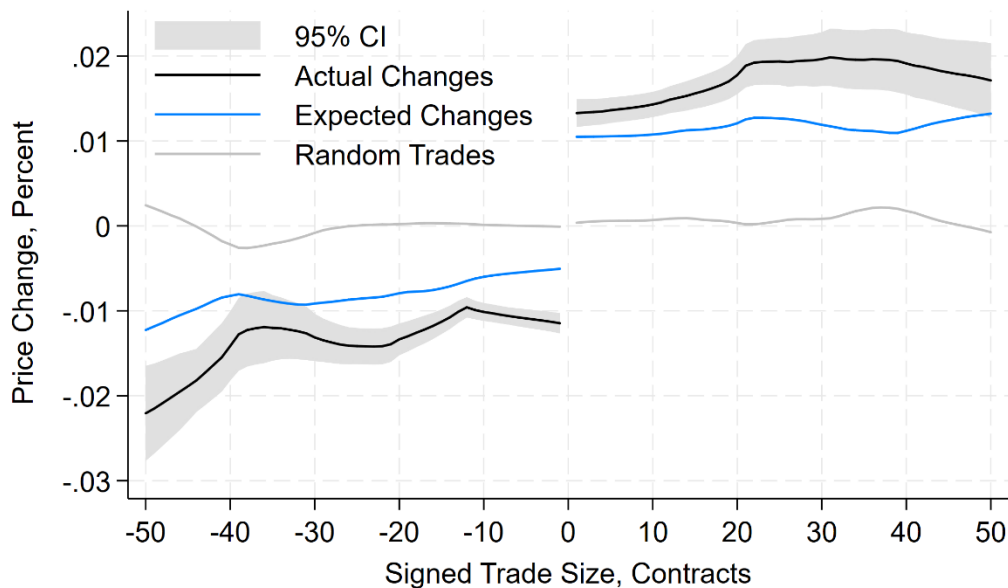


Figure 3

Actual and expected option price changes conditional on signed trade size

The figure shows estimates actual and expected option price changes as a function of signed option trade size. It uses all trades (10,904 in total) for options on Bank of America stock during January 2007. The blue line around 1% shows the predicted change in option mid-price based on pre-trade public information computed using Equation (3). The light grey line around zero shows the option price changes following a set of simulated trades for the same option and date at random times that do not overlap with the ten-minute periods following the time of an actual trade. The black line shows the relative change in the option mid-price in ten-minutes after the trade. Its 95% confidence interval is shown in light grey. All price changes are normalized by the option quote midpoint at the time of the trade and computed over a ten-minute horizon. Signed trade size is based on option trade size in contracts adjusted for trade direction. We use kernel-weighted local polynomial smoothing with the Epanechnikov kernel to estimate the price changes.



**Internet Appendix to
Options Trading Costs Are Lower Than You Think**

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This Internet Appendix provides additional empirical results that supplement those in the main text. Tables IA.1-IA.5, IA.7-IA.8, and IA.10 report results for options on stocks that are not included in the S&P 500 index (non-S&P 500 stocks). They parallel Tables 1-5, 7-8, and 10 in the main text which report results for options on stocks in the S&P 500 index (S&P 500 stocks).

Table IA.6 reports estimates of options trading costs and price impact for portfolios of options sorted by order imbalances in calls (Panel A) and puts (Panel B). Panel A follows the format of Table 6 in the main text, which reports estimates of trading costs for portfolios sorted by several other variables. Panel B supplements the results in Table 8 of the main text, which reports estimates of price impact for the full sample and for different option moneyness categories.

Table IA.9 reports the results of stock-by-date panel regressions that explore how the fit of the predictive models as measured by R^2 varies with time-series shocks to volatility, returns, and information asymmetry.

Table IA.11 presents summary statistics related to the samples of quotes used to estimate Models 1 and 2. These regression models are estimated on quote snapshots at one-minute frequency, where the left-hand side variable is the change in the midpoint quote over the next ten minutes. Table IA.11 reports statistics about the frequency with which these ten-minute intervals contain an option trade and the frequency that these options trades consume the quoted size at the NBBO.

Table IA.12 reports trading cost estimates separately for calls and puts on S&P 500 stocks, and for options on S&P 500 stocks in different time-to-expiration buckets.

Table IA.13 reports the average correlations between the midpoint quote changes over ten-minute horizons, the price changes predicted by Model 1, and the price changes predicted by Model 1, for both S&P 500 and non-S&P 500 stocks.

Table IA.14 reports the results of stock-by-date panel regressions for both S&P 500 and non-S&P 500 stocks that explore how the measures of price impact vary with proxies for volatility and the risk of options market maker inventories of options.

Figure IA.1 consists of several panels showing how the measures of trading costs of different groups of options evolved over time. Figure IA.2 consists of several panels showing

how the R^2 s of Models 1 and 2 varied throughout the sample period. Figure IA.3 shows how the coefficients on the key predictive variable $\hat{P}_t^{BSM} - P_t$ evolved over time.

Table IA.1**Summary statistics for option trades (non-S&P 500 stocks)**

The sample consists of all trades in options on non-S&P 500 stocks from January 2004 through December 2015. Each statistic is computed for each sample month, and the table reports the average values of the statistics across the 144 sample months. Days-to-expiration is in calendar days. Trade price, trade size in number of contracts traded, and stock price are computed immediately prior to the option trade. Market capitalization of the underlying stock is as-of the previous day's close. Minute of day is the time at which the trade occurs, measured from 9:30 a.m. eastern time. Size quoted at ask and bid are the numbers of contracts quoted at the national best ask and bid immediately prior to the option trade. The percentages and numbers of options exchanges quoting at the ask and bid are also measured immediately prior to the trade. The sample contains a total of 202,333,269 options trades in non-S&P 500 stocks, or an average of 1,405,092 trades per month.

Variable	Mean	Std.dev.	Percentiles						
			1%	5%	25%	median	75%	95%	99%
<u>Statistics of options trades</u>									
Call/put (call = 0, put = 1)	0.37	0.48	0	0	0	0	1	1	1
Buy/sell (buy = 1, sell = −1)	−0.036	1.00	−1	−1	−1	−0.9583	1	1	1
Delta	0.46	0.21	0.08	0.14	0.30	0.44	0.59	0.84	0.92
Days-to-expiration	76.68	107.19	1.83	5.34	19.28	37.46	94.32	264.93	601.07
Trade price (\$)	3.04	4.46	0.18	0.32	0.83	1.67	3.44	10.23	21.27
Trade size (contracts)	15.48	57.92	1.00	1.00	2.03	6.56	11.83	48.11	158.03
Stock price (\$)	52.77	52.30	3.20	6.28	18.36	33.77	64.83	163.66	206.28
Market capitalization (\$million)	8,502	12,075	196	444	1,445	3,405	10,551	31,989	53,143
Minute of day (9:30 a.m. ET = 0)	170.3	117.9	4.75	12.51	61.84	152.55	277.10	364.79	377.92
<u>Market condition at time of trade</u>									
Size quoted at ask	594.2	1,401.9	1.0	2.6	25.3	140.6	527.6	2,789.8	6,847.2
Size quoted at bid	630.3	1,449.9	1.0	3.0	28.7	153.5	568.1	2,958.7	7,155.4
% of options exchanges at ask	59.9%	33.1%	13.1%	13.3%	24.4%	67.2%	92.1%	100.0%	100.0%
% of options exchanges at bid	60.8%	33.0%	13.1%	13.4%	25.6%	68.5%	92.6%	100.0%	100.0%
# of options exchanges at ask	4.14	2.52	1	1	1.31	4.38	6.41	7.77	8.21
# of options exchanges at bid	4.20	2.51	1	1	1.46	4.47	6.43	7.79	8.20
<u>Changes in options prices following trades</u>									
Actual change in quote midpoint	0.1%	5.3%	−15.0%	−8.8%	−2.3%	0.0%	2.4%	9.4%	15.7%
Change predicted by Model 1	0.1%	1.5%	−4.4%	−2.3%	−0.6%	0.0%	0.6%	2.5%	4.8%
Change predicted by Model 2	0.1%	2.4%	−6.6%	−3.7%	−1.0%	0.0%	1.1%	4.2%	7.1%

Table IA.2**Conventional measures of option trading costs (non-S&P 500 stocks)**

The sample consists of all trades in options on non-S&P 500 stocks from January 2004 through December 2015. Each statistic is computed for each combination of underlying stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while Panel B uses only trades that fall in a particular moneyness category, which are defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The quoted half-spread is half the difference between the ask and bid prices at the time of the trade, expressed as either a percentage of the pre-trade bid-ask midpoint or in dollars. The effective half-spread is the difference between the trade price and the bid-ask midpoint (bid-ask midpoint and trade price) for trades signed as buys (sells), where a trade is signed as a buy (sell) if the trade price is greater (less) than the midpoint.

Panel A: Options on non-S&P500 stocks

Variable	Mean	Std.dev.	Median
Quoted spread (%)	12.8%	6.7%	11.3%
Effective spread (%)	9.7%	5.6%	8.5%
Price impact (%)	2.2%	2.4%	1.9%
Quoted spread (\$)	0.20	0.22	0.15
Effective spread (\$)	0.15	0.14	0.11
Number of stock-days	2,440,621		

Panel B: Options split by moneyness

	<u>OTM</u>		<u>ATM</u>		<u>ITM</u>	
	<u>Mean</u>	Std.dev.	<u>Mean</u>	Std.dev.	<u>Mean</u>	Std.dev.
Quoted spread (%)	16.6%	8.1%	11.6%	7.0%	8.4%	6.1%
Effective spread (%)	12.9%	7.2%	8.8%	5.8%	6.1%	4.7%
Price impact (%)	2.7%	3.2%	2.1%	2.6%	1.6%	2.4%
Quoted spread (\$)	0.14	0.13	0.19	0.20	0.29	0.33
Effective spread (\$)	0.10	0.09	0.14	0.14	0.20	0.22
Num. of stock-days	1,496,725		2,055,289		1,659,885	

Table IA.3**Predictive model estimates (non-S&P 500 stocks)**

Coefficient estimates and t -statistics for the predictive Models 1 and 2 in Equations (2) and (3) estimated using options on non-S&P 500 stocks. Model 1 predicts ten-minute changes in the option bid-ask midpoint $P_{t+\tau} - P_t$ using $\hat{P}_t^{BSM} - P_t$, the difference between the BSM value and the time t option quote midpoint. Model 2 also uses $\%ExchBid_t$ and $\%ExchAsk_t$, the percentages of options exchanges with bid or ask price equal to the National Best Bid (Offer), $\Delta \times (S_{t-(j-1)\Delta t} - S_{t-j\Delta t})$ for $j = 1$ and 2 , the two most recent lagged one-minute delta-adjusted changes in the underlying stock price, and $P_{t-(j-1)\Delta t} - P_{t-j\Delta t}$, for $j = 1$ and 2 , the two most recent lagged changes in the option quote midpoint. Moneyness categories are defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. We first estimate a separate regression for each option contract and day using snapshots of options quotes at one-minute frequency, without using any trade data. We then compute the average coefficient estimates for each stock and month and report the averages over the pooled stock-by-month panel. t -statistics based on robust standard errors clustered by stock and month are in parentheses below the coefficient estimates. The sample consists of options on non-S&P 500 stocks during the period from January 2004 through December 2015.

	Model 1			Model 2		
	OTM	ATM	ITM	OTM	ATM	ITM
Constant	0.0007 (17.50)	0.0002 (6.67)	-0.0026 (-28.89)	-0.0004 (-8.00)	-0.0001 (-4.17)	-0.0043 (-15.25)
Implied bias	0.3287 (63.46)	0.3903 (59.77)	0.4438 (48.03)	0.2428 (56.73)	0.37 (51.85)	0.4682 (41.22)
$\%ExchBid_t$				0.0174 (46.58)	0.025 (35.13)	0.0383 (27.42)
$\%ExchAsk_t$				-0.0166 (-36.60)	-0.025 (-33.29)	-0.0354 (-29.84)
$\Delta \times (S_t - S_{t-\Delta t})$				0.1233 (70.94)	0.1338 (62.29)	0.0887 (50.11)
$\Delta \times (S_{t-\Delta t} - S_{t-2\Delta t})$				0.0482 (33.68)	0.0493 (31.58)	0.0267 (21.68)
$P_t - P_{t-\Delta t}$				-0.0461 (-17.25)	-0.0546 (-10.41)	-0.0209 (-3.13)
$P_{t-\Delta t} - P_{t-2\Delta t}$				-0.0127 (-7.80)	-0.0178 (-0.64)	-0.0029 (-4.62)
Adj. R^2	0.0842	0.0823	0.0895	0.1693	0.1340	0.1295
N	271,928	324,626	330,512	228,756	266,102	268,319

Table IA.4.**Predictive model out-of-sample performance (non-S&P 500 stocks)**

Measures of the out-of-sample performance of the predictive models. The results for the out-of-sample R^2 s in the first row of results are based the formula for the out-of-sample R^2 in Equation (4). For each month, we estimate the R^2 s using all option trades on non-S&P 500 stocks during the month and then report the mean, median, and standard deviation of the monthly R^2 s. The last three rows report results from no-intercept regressions of changes in mid-quote prices on the ex-ante forecasts from each model. We estimate regression separately for each month and report the statistics for the time series of monthly estimates. In both cases, the Model 1 and 2 forecasts are computed using the pre-trade values of the covariates and the coefficient estimates from Equations (2) and (3) estimated using data from the previous month.

Parameter	Model 1			Model 2		
	Mean	Std.dev.	Median	Mean	Std.dev.	Median
Out-of-sample R^2	9.7%	2.9%	9.1%	12.3%	4.2%	12.5%
<u>Regressions of realized price change on forecast</u>						
b	1.08	0.14	1.09	0.80	0.08	0.83
t -statistic	(369.08)	(62.16)	(376.76)	(440.07)	(76.01)	(448.67)
R^2	9.8%	3.0%	9.2%	13.1%	3.6%	12.9%
No. obs. each month	1,399,314	571,329	1,346,300	1,393,738	569,679	1,338,997

Table IA.5.**Trading costs estimates (non-S&P 500 stocks)**

The sample consists of all trades in options on non-S&P 500 stocks from January 2004 through December 2015. Each statistic is computed for each combination of stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while Panel B are for trades in the given moneyness category defined by the absolute value of the option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The quoted half-spread is half the difference between the ask and bid prices at the time of the trade, expressed as either a percentage of the pre-trade bid-ask midpoint or in dollars. The effective half-spread is the difference between the trade price and the bid-ask midpoint (bid-ask midpoint and trade price) for trades signed as buys (sells). The adjusted effective half-spread is computed using Equation (7) and represents average costs across execution timers and non-timers. The cost for execution timers (the algo spread) is computed using Equation (12). The execution timing share is the fraction of trades that show timing and is computed using Equation (6).

. Panel A: All options

Variable	Mean	Std.dev.	Median
Quoted half-spread (%)	12.8%	6.7%	11.3%
Effective half-spread (%)	9.7%	5.6%	8.5%
Adjusted half-spread (%)	7.7%	6.1%	6.4%
Algo. half-spread (%)	5.4%	8.1%	4.0%
Quoted half-spread (\$)	0.20	0.22	0.15
Effective half-spread (\$)	0.15	0.14	0.11
Adjusted half-spread (\$)	0.11	0.14	0.08
Algo. half-spread (\$)	0.07	0.14	0.05
Execution timing share (%)	37.3%	50.5%	42.9%
Number of stock-days	2,440,621		

Panel B: Averages by moneyness category

Variable	OTM	ATM	ITM
Quoted half-spread (%)	16.6%	11.6%	8.4%
Effective half-spread (%)	12.9%	8.8%	6.1%
Adjusted half-spread (%)	10.6%	6.9%	4.4%
Algo. half-spread (%)	8.0%	4.7%	2.8%
Quoted half-spread (\$)	0.14	0.19	0.29
Effective half-spread (\$)	0.10	0.14	0.20
Adjusted half-spread (\$)	0.09	0.11	0.14
Algo. half-spread (\$)	0.06	0.07	0.08
Execution timing share (%)	38.7%	40.0%	42.5%
Number of stock-days	1,496,725	2,055,289	1,659,885

Table IA.6.**Options trading costs for various groups of stocks**

This table reports results for two additional sets of sorted portfolios based on call and put order imbalances for which results are not reported in Tables 6 and 8. Each day, we sort stocks into five portfolios using the order imbalances in either calls or puts over the previous 21 trading day divided by option trading volume over the same period. The table reports the quoted half-spread, the conventional effective half-spreads, the adjusted effective half-spread, and the algo. half-spread. We first compute the portfolio averages for each day and then report the overall averages across days. The last two columns report the fraction of trades with execution timing and the average values of the variable used to sort the stocks into portfolios. The sample consists of options on all common stocks (not just S&P 500 stocks) during the period from January 2004 through December 2015, and thus each portfolio has slightly more than 194 stocks. The last row of each panel reports the difference between portfolios 5 and 1. All averages reported in the table are statistically significant due to the large sample sizes.

Panel A. Order imbalances for calls

Portfolio	Trading cost measures (half-spreads)				Exec. Timing %	Price impact		Order imbalance
	Quoted spread	Effective spread	Adj. eff. spread	Algo. spread		Conventional	Adjusted	
1	0.107	0.086	0.065	0.041	42%	0.021	0.010	-0.267
2	0.091	0.074	0.054	0.029	40%	0.018	0.008	-0.070
3	0.086	0.071	0.052	0.027	39%	0.017	0.008	0.003
4	0.095	0.078	0.057	0.032	40%	0.019	0.009	0.080
5	0.112	0.091	0.068	0.045	42%	0.022	0.011	0.283
Diff	0.0047	0.0041	0.0037	0.004	-0.17%	0.0009	0.0007	0.5503
	[7.6]	[12.3]	[12.1]	[12.3]	[-0.9]	[7.2]	[9.2]	[304.4]

Panel B. Order imbalances for puts

Portfolio	Trading cost measures (half-spreads)				Exec. Timing %	Price impact		Order imbalance
	Quoted spread	Effective spread	Adj. eff. spread	Algo spread		Conventional	Adjusted	
1	0.112	0.091	0.069	0.045	41%	0.021	0.011	-0.351
2	0.092	0.075	0.055	0.029	41%	0.018	0.008	-0.086
3	0.085	0.070	0.051	0.026	40%	0.017	0.008	0.009
4	0.092	0.075	0.055	0.030	41%	0.018	0.008	0.110
5	0.109	0.089	0.067	0.043	42%	0.021	0.010	0.371
Diff	-0.0026	-0.002	-0.0022	-0.0023	0.0047	-0.0003	-0.0003	0.7228
	[-4.2]	[-6.1]	[-7.4]	[-7.1]	[2.5]	[-2.2]	[-4.2]	[342.7]

Table IA.7**Time series variation in options trading costs (non-S&P 500 stocks)**

Regression results showing how options trading costs respond to time-series shocks to stock volatility and information asymmetry. We estimate stock-day panel regression with stock fixed effects and dependent variables consisting of relative conventional and algo effective half-spreads, their difference, and the execution timing share. The independent variables include short and mid-term volatility measures: the absolute abnormal stock return from the previous day ($|StkRet_{t-1}|$) and the standard deviation of abnormal stock returns in the previous 21 days ($Std(AR)_{1m}$), and measures of asymmetric information: indicator variables for the earnings and pre-earnings announcement days (EAD_t, EAD_{t-1}). The panel include 2,440,507 stock-days. Standard errors are clustered by stock and by day.

	Conv. half-spread	Algo half-spread	Difference	Exec. timing share
$ Ret_{t-1} $	0.028 [9.3]	-0.04 [-11.1]	0.068 [26.3]	0.2015 [10.7]
$Std(AR)_{1m}$	0.0045 [15.8]	-0.0135 [-29.6]	0.018 [48.9]	-0.0156 [-6.4]
EAD_t	0.0036 [11.0]	-0.0028 [-6.5]	0.0064 [19.7]	-0.0211 [-8.1]
EAD_{t-1}	0.1136 [10.7]	0.059 [6.1]	0.0546 [13.2]	0.2457 [6.4]
R^2	12%	8%	1%	1%

Table IA.8**Estimates of price impact (non-S&P 500 stocks)**

Conventional and adjusted measures of price impact. The price impact of a buy trade is the difference between the midpoint ten minutes after the trade and the pre-trade midpoint, expressed as a percentage of the pre-trade midpoint, or the negative of this for a sell trade. Adjusted price impacts are computed using Equation (9) by subtracting the expected change in the option price computed using Model 1 or 2 from the actual price change during the ten minute period following a trade. Each statistic is computed for each combination of stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A uses all option trades, while each set of results in Panel B uses only trades for the indicated moneyness category, which is defined by absolute option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM options. The sample consists of all trades in options on S&P 500 stocks during the period from January 2004 through December 2015.

Panel A: All options

Variable	Mean	Std.dev.	Median
Conventional measure	2.225%	2.418%	1.865%
Adjusted using Model 1	1.561%	2.137%	1.257%
Adjusted using Model 2	1.233%	2.124%	0.914%
No. of stock-days	2,440,621		

Panel B: Averages by moneyness category

	OTM	ATM	ITM
Conventional measure	2.655%	2.084%	1.580%
Adjusted using Model 1	1.951%	1.418%	0.966%
Adjusted using Model 2	1.530%	1.100%	0.709%
No. of stock-days	1,496,725	2,055,289	1,659,885

Table IA.9.**Time series variation in model R^2 s**

This table reports regression results showing how the fit of the predictive models as measured by R^2 depends on time-series shocks to stock volatility, returns, and information asymmetry. We estimate stock-day panel regression with stock fixed effects and the R^2 s of Models 1 and 2 using samples of options on S&P 500 and non-S&P 500 stocks. The independent variables include absolute abnormal stock return in the previous day ($|Ret_{t-1}|$), standard deviation of abnormal stock returns in the previous 21 days ($Std(AR)_{1m}$), indicator variables for the earnings and pre-earnings announcement days (EAD_t, EAD_{t-1}), the previous-day stock return, and stock market capitalization in millions of dollars. Standard errors are clustered by stock and by day. Results are reported separately for Models 1 and 2 and for options on S&P 500 and non-S&P 500 stocks.

	<u>S&P 500 stocks</u>		<u>Non-S&P 500 stocks</u>	
	R^2 for Model 1	R^2 for Model 2	R^2 for Model 1	R^2 for Model 2
Market Cap	-0.00 [-1.3]	-0.00 [-0.2]	-0.00 [-5.0]	-0.00 [-4.6]
$Std(AR)_{1m}$	-0.0734 [-3.9]	-0.145 [-4.0]	-0.0408 [-7.7]	-0.0922 [-7.8]
EAD_t	-0.0121 [-24.2]	-0.0118 [-15.4]	-0.0068 [-19.8]	-0.0108 [-18.3]
EAD_{t-1}	-0.006 [-16.8]	-0.0107 [-18.5]	-0.0045 [-15.3]	-0.012 [-22.3]
Ret_{t-1}	0.0195 [3.6]	0.0321 [3.5]	-0.0004 [-0.2]	0.0075 [1.9]
$ Ret_{t-1} $	-0.0815 [-11.0]	-0.1557 [-12.3]	0.0005 [0.2]	-0.0594 [-12.5]
R^2	0.12	0.24	0.04	0.1
N	1,324,195	1,322,223	3,982,003	3,619,382

Table IA.10**Impact of execution timing on the after-cost profitability of options trading strategies (non-S&P 500 stocks)**

Returns from selling straddles with several assumptions about transactions costs. On the trading day following the expiration of options on the regular cycle, we buy a straddle consisting of a call and a put with the same strike and expiration in about one month and hold the straddle until it expires. We compute straddle returns following Equation (10). We form ten portfolios each month and report portfolio average return over all months. Panels A and B report the equally-weighted average returns of portfolios sorted by the difference between historical and implied volatility ($HV - IV$) and by the previous month's stock return, respectively. The first row of results reports straddle returns are assuming that trades are at mid-quote prices, and thus assumes that there are no transactions costs. The next two rows adjust returns using estimates of conventional and algo. effective half-spreads by adding the half-spreads to the no transactions costs straddle returns. The last three rows of the table report the estimates of the transactions costs and the average values of the sorting variable ($HV - IV$ or the previous month's stock return) for the portfolios. The t -statistics are based on standard errors adjusted for heteroscedasticity and autocorrelation. The sample period runs from January 2004 through December 2015.

Panel A: Portfolios sorted on $HV - IV$

Portfolio:	L= 1	2	3	4	5	6	7	8	9	H = 10	H - L
Straddle return:											
No transactions costs	-0.089*** [-6.1]	-0.043** [-2.2]	-0.03 [-1.4]	-0.008 [-0.4]	-0.022 [-0.9]	-0.011 [-0.5]	-0.005 [-0.2]	-0.011 [-0.5]	-0.017 [-0.7]	-0.002 [-0.1]	0.087*** [6.5]
Algo eff. half-spread	-0.041*** [-2.9]	-0.001 [-0.0]									
Conven. eff. half-spread	-0.014 [-1.0]	0.027 [1.4]									
Other statistics:											
Conv. eff. half-spread	0.075	0.07	0.069	0.067	0.067	0.068	0.068	0.068	0.07	0.076	0.001
Algo eff. half-spread	0.048	0.042	0.042	0.04	0.039	0.04	0.04	0.04	0.042	0.047	-0.001
$HV - IV$	-0.194	-0.067	-0.037	-0.018	-0.002	0.014	0.031	0.052	0.084	0.205	0.399

Panel B: Portfolios sorted on previous month's stock return

Portfolio:	L= 1	2	3	4	5	6	7	8	9	H = 10	H - L
Straddle return:											
No transactions costs	-0.037*	-0.012	-0.005	-0.014	-0.004	-0.007	-0.021	-0.033	-0.043**	-0.061***	-0.025
	[-1.7]	[-0.5]	[-0.2]	[-0.6]	[-0.2]	[-0.3]	[-1.0]	[-1.6]	[-2.1]	[-3.7]	[-1.4]
Algo eff. half-spread									-0.001	-0.017	
									[-0.1]	[-1.0]	
Conven. eff. half-spread									0.026	0.01	
									[1.3]	[0.6]	
Other statistics:											
Conv. eff. half-spread	0.076	0.071	0.07	0.069	0.068	0.067	0.068	0.067	0.069	0.072	-0.004
Algo eff. half-spread	0.046	0.043	0.042	0.041	0.041	0.039	0.041	0.04	0.041	0.044	-0.002
Previous month's stock return	-0.204	-0.087	-0.049	-0.024	-0.002	0.017	0.038	0.063	0.098	0.202	0.406

Table IA.11**Frequency of options trades in the quote sample used to estimate the predictive models**

The predictive models estimates changes in the midpoint over ten-minute horizons. We estimate the models for each option contract and day using quote snapshots at one-minute frequency. We then average the coefficients across all options with given moneyness for a stock-day. For S&P 500 stocks, a typical stock-day includes 12.9 ATM option contracts, and these ATM options have 238 trades in total. The table provides information about the number of ten-minute intervals that contain trades, and the fraction of trades that consume the bid or ask quoted at the time of the trade. A typical ATM contract only has 3.9 ten-minute intervals per day that contain any trades. Conditional on a trade, only 22% of the trades consume the quoted size at the NBBO. A typical ATM option trade is about 34% of the quoted size at the reporting exchange.

Panel A: S&P 500 stocks	OTM options		ATM options		ITM options	
	Mean	Median	Mean	Median	Mean	Median
Number of options used in the model estimation, per stock day	15.21	8.09	12.94	8.45	10.08	5.48
Number of trades	219.73	42.00	238.19	49.00	89.44	19.00
Number of the 10-minute intervals that contain trades, per stock day	2.97	2.43	3.94	3.06	2.85	2.29
Fraction of trades that consume the NBBO bid or ask size at the time of the trade	0.19	0.17	0.22	0.19	0.25	0.21
Ratio of trade size to NBBO quoted size	0.33	0.29	0.34	0.31	0.35	0.31

Panel B: non-S&P 500 stocks	OTM options		ATM options		ITM options	
	Mean	Median	Mean	Median	Mean	Median
Number of options used in the model estimation, per stock day	5.48	1.94	5.30	2.46	3.96	1.92
Number of trades	44.85	9.00	47.01	10.00	21.41	5.00
Number of the 10-minute intervals that contain trades, per stock day	2.17	1.75	2.48	2.00	2.06	1.50
Fraction of trades that consume the NBBO bid or ask size at the time of the trade	0.25	0.19	0.27	0.22	0.29	0.22
Ratio of trade size to NBBO quoted size	0.42	0.33	0.42	0.35	0.41	0.33

Table IA.12**Trading cost estimates by option characteristics**

The sample consists of all trades in options on S&P 500 stocks between January 2004 and December 2015. Each statistic is computed for each combination of stock and trade date, and the table reports the average values of the statistics across the stock-dates. Panel A reports results for call and put trades separately, while Panel B reports results for trades grouped by time to maturity measured in trading days. Short-term options expire within 5 to 21 trading days, mid-term options within 22 to 63 days, and long-term options within 64 to 251 days.

Panel A: Call or put

Variable	Calls			Puts		
	Mean	Std.dev.	Median	Mean	Std.dev.	Median
Quoted half-spread (%)	12.8%	7.3%	11.4%	10.6%	6.5%	9.2%
Effective half-spread (%)	10.4%	6.4%	9.0%	8.3%	5.4%	7.1%
Adjusted spread (%)	8.2%	6.7%	6.5%	6.3%	5.5%	4.9%
Algo half-spread (%)	5.4%	8.4%	3.8%	4.1%	7.0%	2.8%
Quoted half-spread (\$)	0.088	0.092	0.069	0.095	0.105	0.071
Effective half-spread (\$)	0.068	0.062	0.055	0.072	0.070	0.056
Adjusted half-spread (\$)	0.053	0.058	0.041	0.054	0.065	0.039
Algo half-spread (\$)	0.034	0.062	0.022	0.035	0.065	0.023
Execution timing (%)	35.8%	46.2%	39.1%	41.6%	45.8%	45.5%
Number of stock-days	1,029,832			1,030,319		

Panel B: Time-to-expiration

Variable	Short-term			Mid-term			Long-term		
	Mean	Std.dev.	Median	Mean	Std.dev.	Median	Mean	Std.dev.	Median
Quoted half-spread (%)	15.4%	9.1%	13.2%	12.9%	8.0%	11.2%	10.1%	6.9%	8.4%
Effective half-spread (%)	12.1%	8.0%	9.8%	10.1%	7.0%	8.5%	8.2%	5.9%	6.7%
Adjusted half-spread (%)	9.0%	8.3%	6.7%	7.7%	7.2%	5.6%	6.5%	6.0%	4.8%
Algo half-spread (%)	5.1%	10.0%	3.1%	4.8%	8.5%	3.0%	4.6%	7.0%	3.1%
Quoted half-spread (\$)	0.063	0.077	0.050	0.075	0.083	0.060	0.098	0.114	0.071
Effective half-spread (\$)	0.046	0.046	0.041	0.056	0.051	0.050	0.076	0.078	0.057
Adjusted half-spread (\$)	0.032	0.039	0.023	0.040	0.046	0.031	0.058	0.070	0.042
Algo half-spread (\$)	0.015	0.047	0.009	0.022	0.054	0.015	0.040	0.071	0.026
Execution timing (%)	37.2%	48.5%	40.2%	39.0%	46.8%	42.9%	39.5%	48.9%	44.7%
Number of stock-days	392,346			921,205			1,016,382		

Table IA.13**Correlations between actual and predicted price changes**

The sample consists of all option trades between January 2004 and December 2015. Each correlation is computed for all trades in each calendar month and the table reports the average values of the statistics across months. Actual change refers to change in option quote midpoint from immediately before a trade to ten minutes after a trade. Predicted changes for Models 1 and 2 for the same trade are computed combining pre-trade information with model coefficients estimated on the historical data. Correlations are reported separately for S&P 500 and non S&P 500 stocks.

Panel A: Options on S&P 500 stocks

Variable	Actual change in option midpoint	Change predicted by Model 1	Change predicted by Model 2
Actual change in option midpoint	100%	22%	28%
Change predicted by Model 1	22%	100%	70%
Change predicted by Model 2	28%	70%	100%

Panel B: Options on non-S&P 500 stocks

Variable	Actual change in option midpoint	Change predicted by Model 1	Change predicted by Model 2
Actual change in option midpoint	100%	31%	36%
Change predicted by Model 1	31%	100%	73%
Change predicted by Model 2	36%	73%	100%

Table IA.14**Relation between price impact and measures of uncertainty and inventory risk**

Results of regressions relating estimates of price impact to covariates that are proxies for uncertainty and inventory risk. The measures of price impact are the conventional measure defined as the difference between the quote midpoint 10 minutes after the trade and the quote midpoint at the time of the trade, and the adjusted measure defined as the difference between the quote midpoint 10 minutes after the trade and the estimate of fair value at the time of the trade computed using Model 2. The covariates include the absolute abnormal stock return on the previous day $|Ret_{t-1}|$, the standard deviation of abnormal stock returns over the previous 21 days $Std(AR)_{1m}$, indicator variables for the earnings and pre-earnings announcement days (EAD_t, EAD_{t-1}), and the absolute value of inventory risk for calls and puts, where inventory risk is computed as non-market-maker order imbalance divided by total volume in the previous thirty days. All regressions include stock fixed effects. Standard errors are clustered by stock and date.

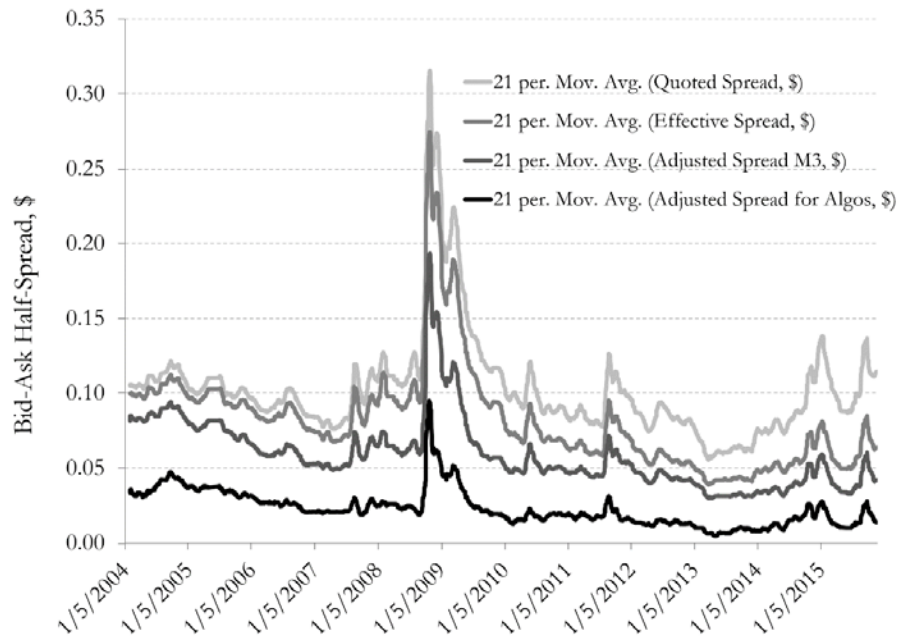
	<u>S&P 500 stocks</u>		<u>Non-S&P 500 stocks</u>	
	Conventional Measure	Based on Model 2	Conventional Measure	Based on Model 2
$Std(AR)_{1m}$,	0.0741	0.0517	0.0418	0.0165
	[13.6]	[10.7]	[8.1]	[8.2]
$ Ret_{t-1} $	0.018	0.0158	0.0189	0.0148
	[10.6]	[12.4]	[16.1]	[19.4]
EAD_t	0.0019	0.0019	0.0023	0.0025
	[13.6]	[18.9]	[13.2]	[23.9]
EAD_{t-1}	0.0017	0.0003	0.0016	0.0004
	[13.6]	[3.3]	[10.0]	[4.0]
$ Inventory_{call} $	0.0009	0.0001	0.0028	0.0004
	[3.3]	[0.4]	[12.5]	[2.9]
$ Inventory_{put} $	0.0007	0.0001	0.0022	0.0004
	[2.9]	[0.9]	[12.6]	[4.7]
R^2	14%	9%	6%	3%
N	743,560	743,560	1,079,483	1,079,483
Fixed Effects	625	625	1877	1,877

Figure IA.1

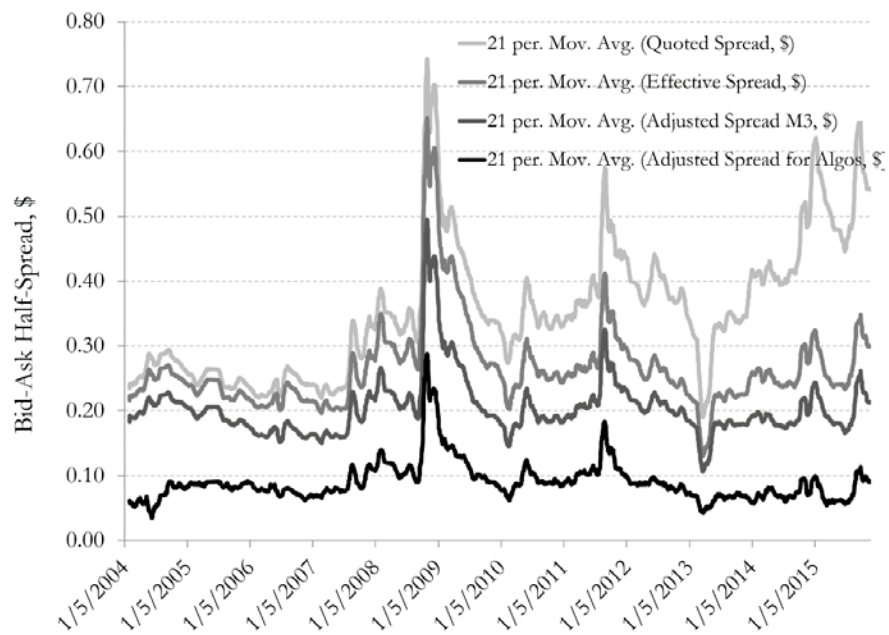
Options trading costs over time

This figure extends Figure 1 and shows the evolution of the average quoted (light grey), the conventionally measured effective (grey), the adjusted effective (dark grey) bid-ask half-spreads and the half-spread for the execution timers (black) over the period from January 2004 to December 2015. We separately report spreads for S&P500 and non-S&P500 stocks, relative and dollar spreads, and spreads for OTM/ATM/ITM trades. Each point on the graph is a 21-day moving average of an average across all option trades on a given day.

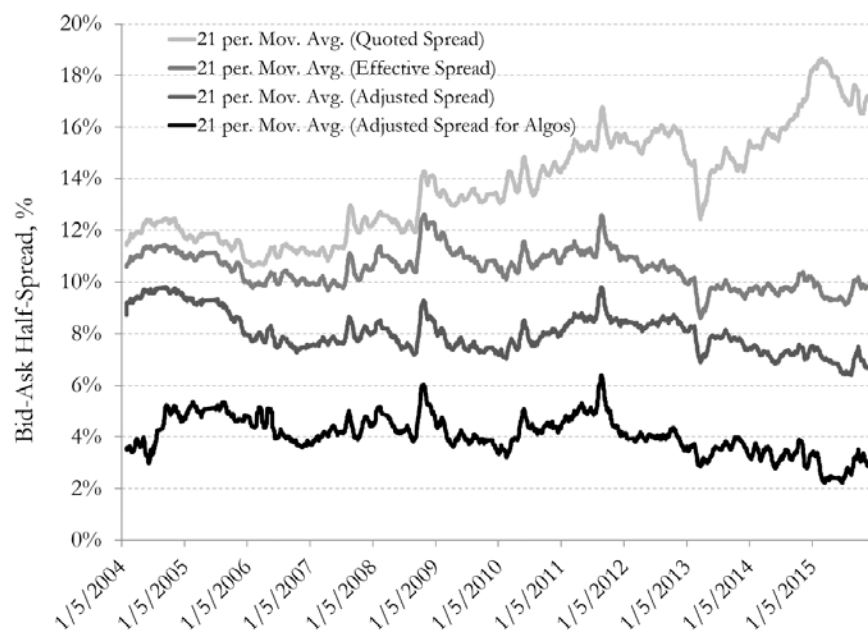
Panel A: Dollar half-spreads of option on S&P500 stocks



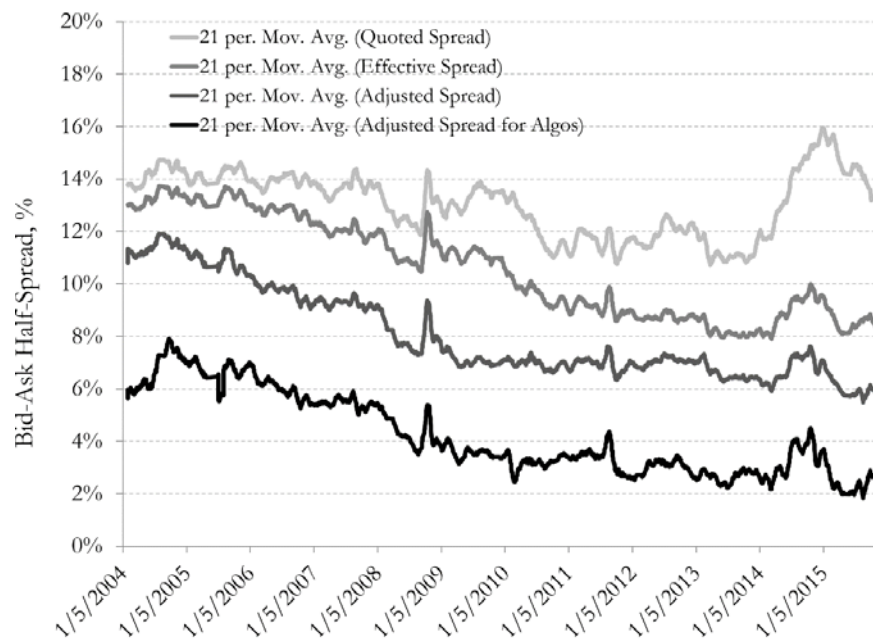
Panel B: Dollar half-spreads of options on non-S&P500 stocks



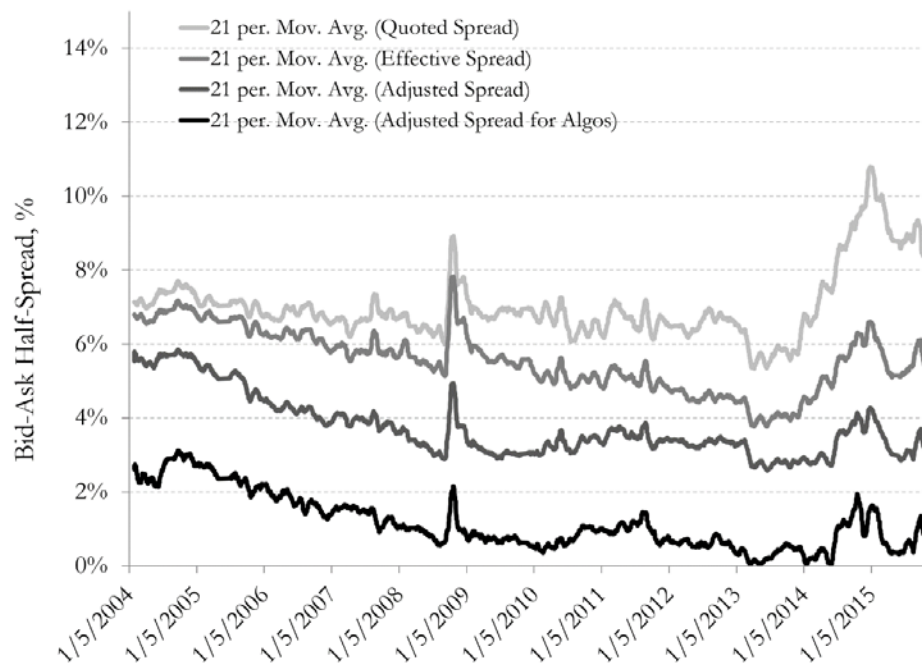
Panel C: Relative half-spreads of options on non-S&P 500 stocks



Panel D: Relative half-spreads of OTM options on S&P500 stocks



Panel E: Relative half-spreads of ATM options on S&P500 stocks



Panel F: Relative half-spreads of ITM options on S&P500 stocks

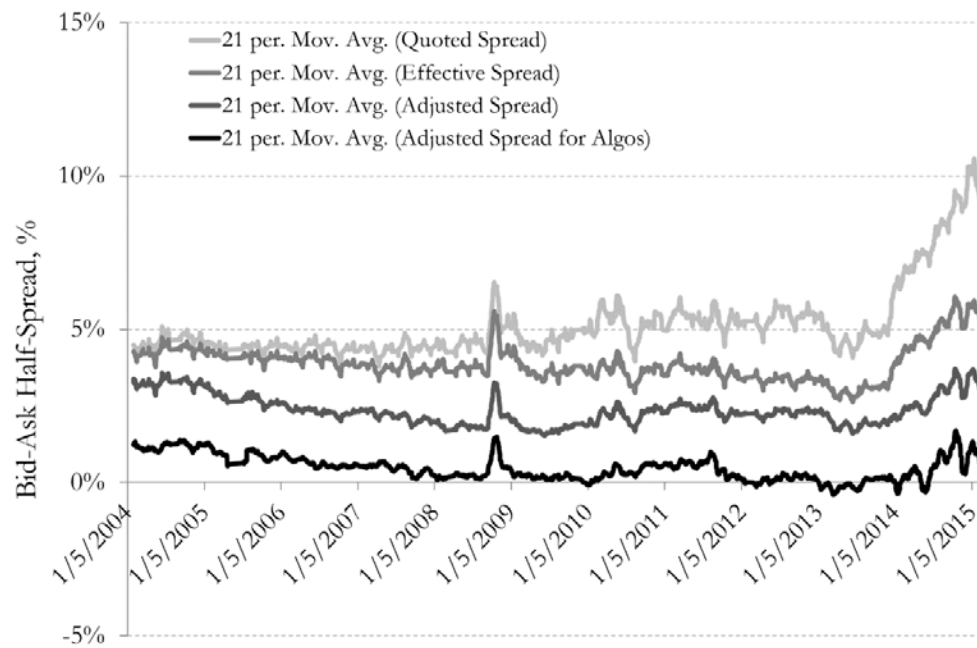


Figure IA.2

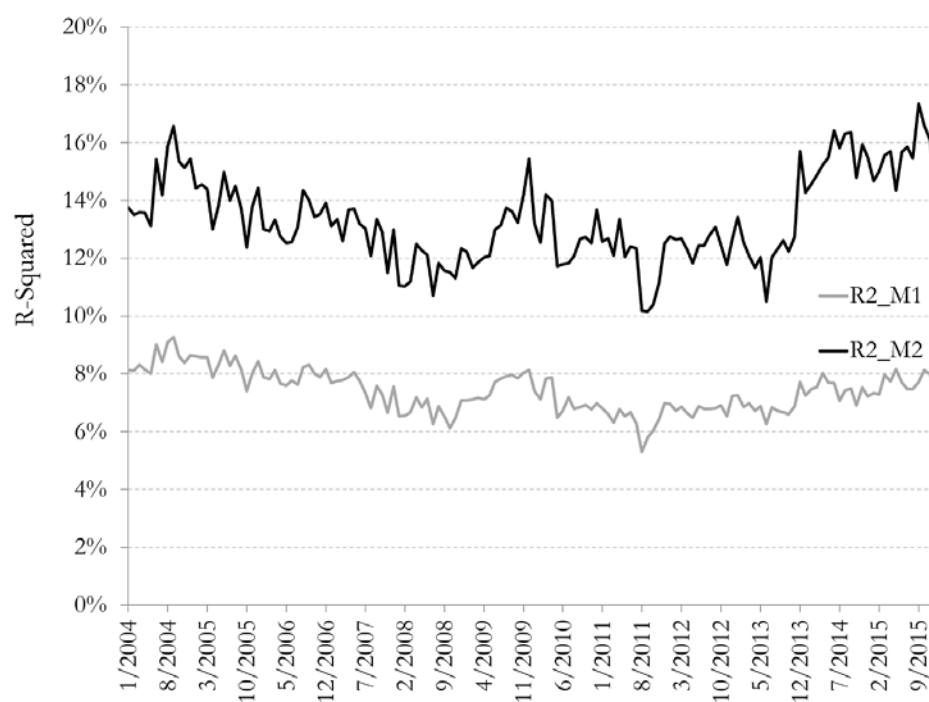
Predictive model R^2 s over time

This figure shows the evolution of the average R^2 in Model 1 (grey) and Model 2 (black) estimated by equations (2) and (3) over the period from January 2004 to December 2015. Each point on the graph is an average over stock-by-day estimates for a given moneyness category in a given month, 144 months in total. Moneyness categories are defined by absolute option delta: $|\Delta| < 0.35$ for OTM, $0.35 \leq |\Delta| < 0.65$ for ATM, and $0.65 \leq |\Delta|$ for ITM.

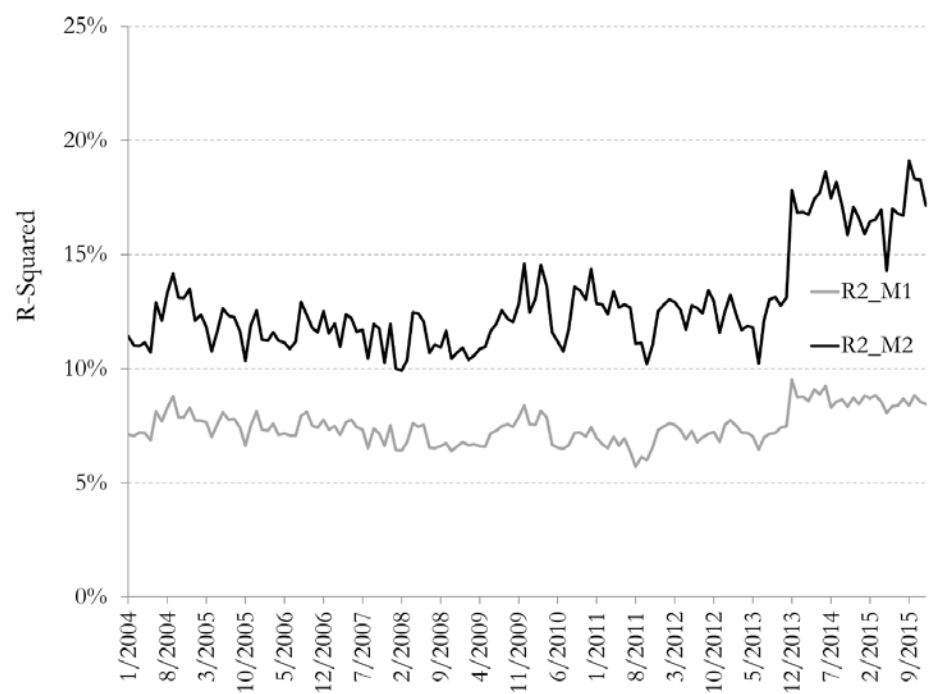
Panel A: R^2 for OTM options on S&P500 stocks



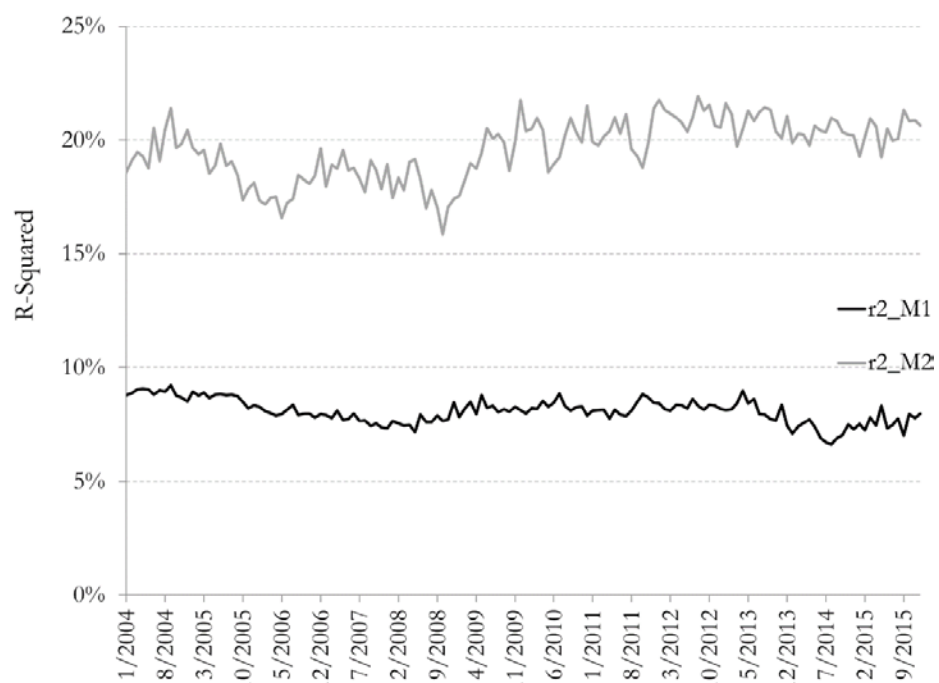
Panel B: R^2 for ATM options on S&P500 stocks



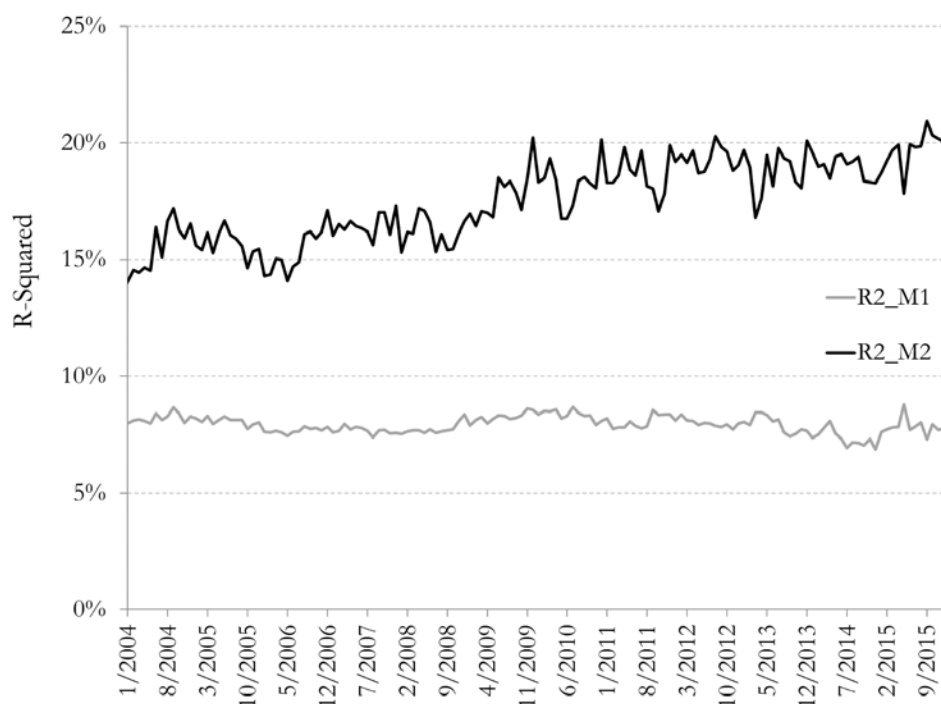
Panel C: R^2 for ITM options on S&P500 stocks



Panel D: R^2 for OTM options on non-S&P500 stocks



Panel E: R^2 for ATM options on non-S&P 500 stocks



Panel F: R^2 for ITM options on non-S&P500 stocks

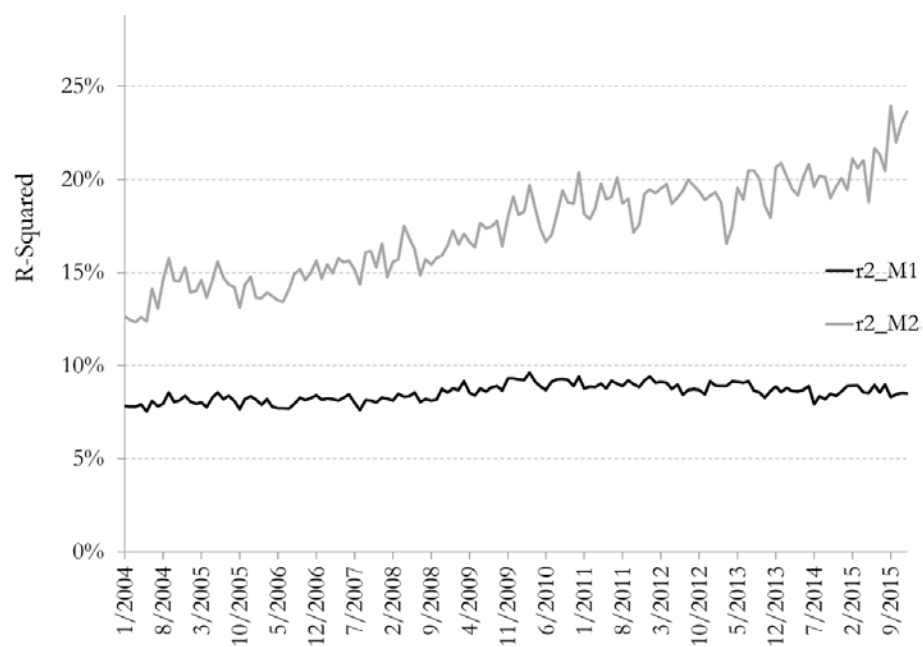
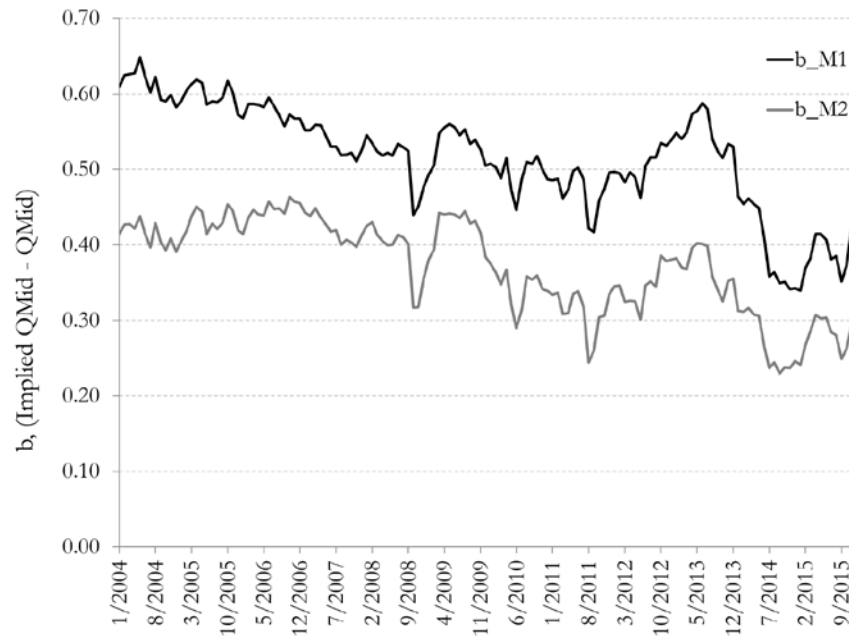


Figure IA.3

Predictive model coefficient estimates over time

Evolution of the coefficient estimates on the predictive variables in Models 1 and 2 estimated using ATM options over the period from January 2004 through December 2015. Each point on the graph is an average over stock-by-day estimates for the ATM options in a given month. ATM options are those for which $0.35 \leq |\Delta| < 0.65$, where Δ denotes the option delta.

Panel A: Coefficients on the key predictive variable $\hat{P}_t^{BSM} - P_t$ for ATM options on S&P 500 stocks



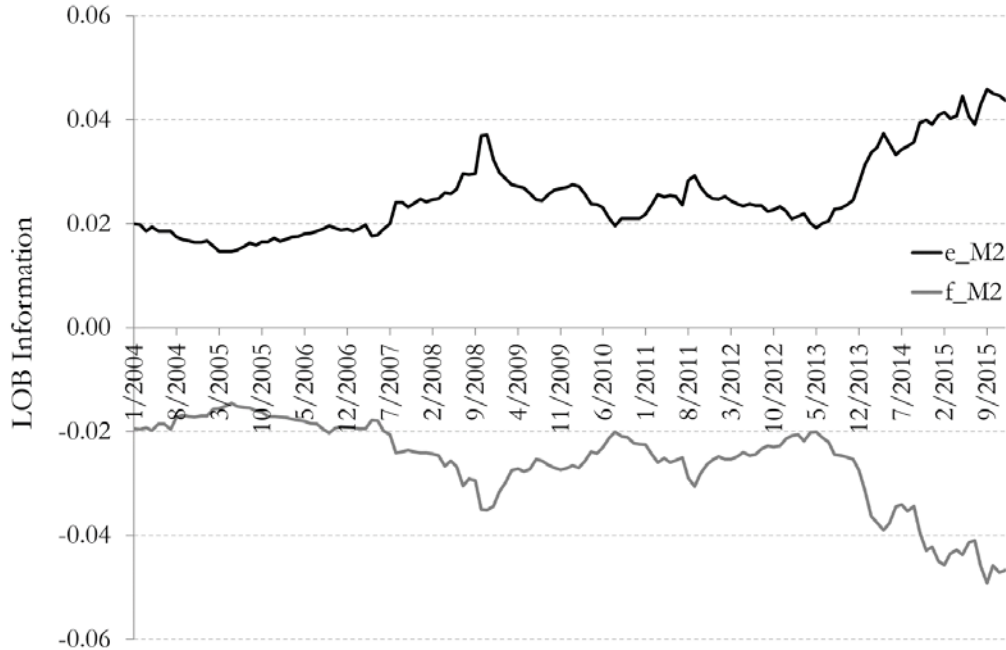
Panel B: Coefficients on the key predictive variable $\hat{P}_t^{BSM} - P_t$ for ATM options on non-S&P 500 stocks



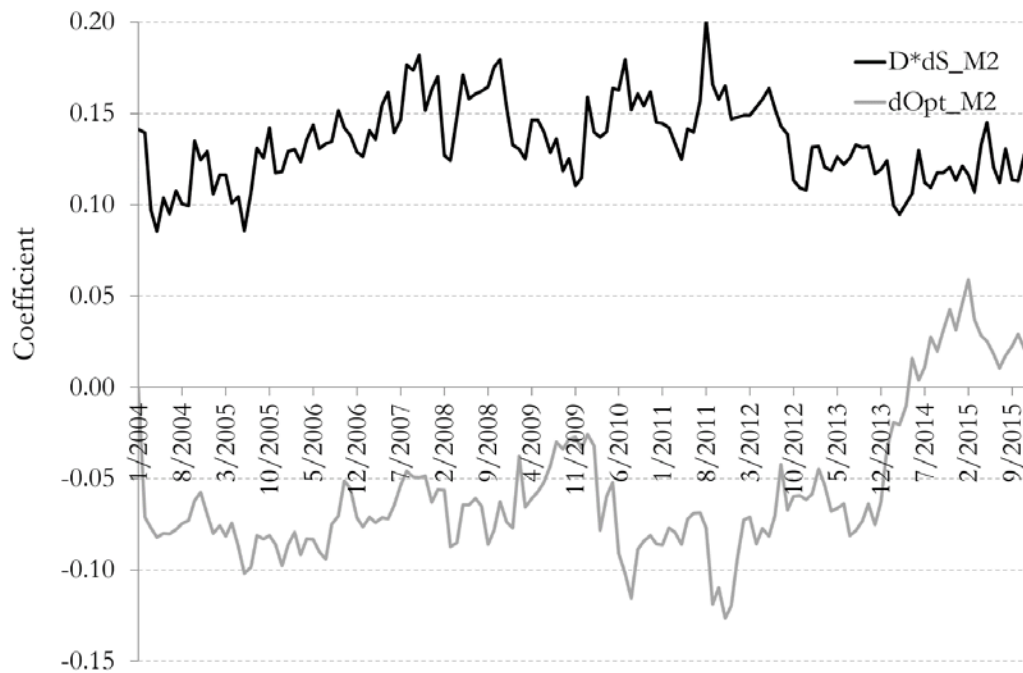
Panel C: Coefficients on the percentages of exchanges at the best bid (black) and ask (grey) prices for ATM options on S&P 500 stocks.



Panel D: Coefficients on the percentages of exchanges at the best bid (black) and ask (grey) prices for ATM options on non-S&P 500 stocks



Panel E: Coefficients on the delta-adjusted stock price (black) and option price (grey) lagged changes for ATM options on S&P 500 stocks.



Panel F: Coefficients on the delta-adjusted stock price (black) and option price (grey) lagged changes for ATM options on non-S&P 500 stocks

